Enterasys® B5

Gigabit Ethernet Switch

Hardware Installation Guide

B5G124-24, B5G124-24P2
B5G124-48, B5G124-48P2
B5K125-24, B5K125-24P2
B5K125-48, B5K125-48P2

P/N 9034527-02



Electrical Hazard: Only qualified personnel should perform installation procedures.

Riesgo Electrico: Solamente personal calificado debe realizar procedimientos de instalacion.

Elektrischer Gefahrenhinweis: Installationen sollten nur durch ausgebildetes und qualifiziertes Personal vorgenommen werden.

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This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

NOTE: This equipment has been tested and found to comply with the limits for a class A digital device, pursuant to Part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment uses, generates, and can radiate radio frequency energy and if not installed in accordance with the operator's manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause interference in which case the user will be required to correct the interference at his own expense.

WARNING: Changes or modifications made to this device which are not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

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This digital apparatus does not exceed the class A limits for radio noise emissions from digital apparatus set out in the Radio Interference Regulations of the Canadian Department of Communications.

Le présent appareil numérique n'émet pas de bruits radioélectriques dépassant les limites applicables aux appareils numériques de la class A prescrites dans le Règlement sur le brouillage radioélectrique édicté par le ministère des Communications du Canada.

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WARNING: This is a Class A product. In a domestic environment this product may cause radio interference in which case the user may be required to take adequate measures.

Clase A. Aviso de ITE

ADVERTENCIA: Este es un producto de Clase A. En un ambiente doméstico este producto puede causar interferencia de radio en cuyo caso puede ser requerido tomar medidas adecuadas.

Klasse A ITE Anmerkung

WARNHINWEIS: Dieses Produkt zählt zur Klasse A (Industriebereich). In Wohnbereichen kann es hierdurch zu Funkstörungen kommen, daher sollten angemessene Vorkehrungen zum Schutz getroffen werden.

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AS/NZS CISPR 22



Hazardous Substances

This product complies with the requirements of European Directive, 2002/95/EC, Restriction of Hazardous Substances (RoHS) in Electrical and Electronic Equipment.

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In accordance with Directive 2002/96/EC of the European Parliament on waste electrical and electronic equipment (WEEE):

1. The symbol above indicates that separate collection of electrical and electronic equipment is required and that this product was placed on the European market after August 13, 2005, the date of enforcement for Directive 2002/96/EC.
2. When this product has reached the end of its serviceable life, it cannot be disposed of as unsorted municipal waste. It must be collected and treated separately.
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4. It is the users' responsibility to utilize the available collection system to ensure WEEE is properly treated.

   For information about the available collection system, please go to www.enterasys.com/services/support/ or contact Enterasys Customer Support at 353 61 705586 (Ireland).

Battery Notice

This product contains a battery used to maintain product information. If the battery should need replacement it must be replaced by Service Personnel. Please contact Technical Support for assistance.



Caution: There is an explosion risk if you replace the battery with the incorrect type. Dispose of expended battery in accordance with local disposal regulations.

Precaución: Hay riesgo de explosion si la bateria se reemplaza con el typo incorrecto. Deshágase de las baterías gastadas de conformidad con las regulaciones de eliminación local.

产品说明书附件
Supplement to Product Instructions

部件名称 (Parts)	有毒有害物质或元素 (Hazardous Substance)					
	铅 (Pb)	汞 (Hg)	镉 (Cd)	六价铬 (Cr^{6+})	多溴联苯 (PBB)	多溴二苯醚 (PBDE)
金属部件 (Metal Parts)	×	○	○	○	○	○
电路模块 (Circuit Modules)	×	○	○	○	○	○
电缆及电缆组件 (Cables & Cable Assemblies)	×	○	○	○	○	○
塑料和聚合物部件 (Plastic and Polymeric parts)	○	○	○	○	○	○
电路开关 (Circuit Breakers)	○	○	○	○	○	○

○： 表示该有毒有害物质在该部件所有均质材料中的含量均在 SJ/T 11363-2006 标准规定的限量要求以下。
Indicates that the concentration of the hazardous substance in all homogeneous materials in the parts is below the relevant threshold of the SJ/T 11363-2006 standard.

×： 表示该有毒有害物质至少在该部件的某一均质材料中的含量超出SJ/T 11363-2006 标准规定的限量要求。
Indicates that the concentration of the hazardous substance of at least one of all homogeneous materials in the parts is above the relevant threshold of the SJ/T 11363-2006 standard.

对销售之日的所售产品, 本表显示,
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此环保使用期限只适用于产品是在产品手册中所规定的条件下工作.
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Safety Information
Class 1 Laser Transceivers

The single mode interface modules use Class 1 laser transceivers.
Read the following safety information before installing or operating these modules.

The Class 1 laser transceivers use an optical feedback loop to maintain Class 1 operation limits. This control loop eliminates the need for maintenance checks or adjustments. The output is factory set, and does not allow any user adjustment. Class 1 Laser transceivers comply with the following safety standards:

- 21 CFR 1040.10 and 1040.11 U.S. Department of Health and Human Services (FDA).
- IEC Publication 825 (International Electrotechnical Commission).
- CENELEC EN 60825 (European Committee for Electrotechnical Standardization).

When operating within their performance limitations, laser transceiver output meets the Class 1 accessible emission limit of all three standards. Class 1 levels of laser radiation are not considered hazardous.

When the connector is in place, all laser radiation remains within the fiber. The maximum amount of radiant power exiting the fiber (under normal conditions) is -12.6 dBm or 55 x 10^{-6} watts.

Removing the optical connector from the transceiver allows laser radiation to emit directly from the optical port. The maximum radiance from the optical port (under worst case conditions) is 0.8 W cm^{-2} or 8 x 10^{3} W m^{2} sr-1.

Do not use optical instruments to view the laser output. The use of optical instruments to view laser output increases eye hazard. When viewing the output optical port, power must be removed from the network adapter.

Safety Compliance

Warning: Fiber Optic Port Safety



When using a fiber optic media expansion module, never look at the transmit laser while it is powered on. Also, never look directly at the fiber TX port and fiber cable ends when they are powered on.

Avertissment: Ports pour fibres optiques - sécurité sur le plan optique



Ne regardez jamais le laser tant qu'il est sous tension. Ne regardez jamais directement le port TX (Tramsmission) à fibres optiques et les embouts de câbles à fibres optiques tant qu'ils sont sous tension.

Warnhinweis: Faseroptikanschlüsse - Optische Sicherheit



Niemals ein Übertragungslaser betrachten, während dieses eingeschaltet ist. Niemals direkt auf den Faser-TX-Anschluß und auf die Faserkabelenden schauen, während diese eingeschaltet sind.

Declaration of Conformity

Application of Council Directive(s):	**2004/108/EC** **2006/95/EC**
Manufacturer's Name:	**Enterasys Networks, Inc.**
Manufacturer's Address:	**50 Minuteman Road** **Andover, MA 01810** **USA**
European Representative Name:	**Enterasys Networks, Ltd.**
European Representative Address:	**Nexus House, Newbury Business Park** **London Road, Newbury** **Berkshire RG14 2PZ, England**
Conformance to Directive(s)/Product Standards:	**EC Directive 2004/108/EC** **EN 55022:2006** **EN 55024:1998** **EN 61000-3-2:2006** **EN 61000-3-3:1995** **EC Directive 2006/95/EC** **EN 60950-1:2006** **EN 60825-1:2007** **EN 60825-2:2004**
Equipment Type/Environment:	**Information Technology Equipment, for use in a Commercial or Light Industrial Environment.**

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About This Guide

This guide provides an overview, installation and troubleshooting instructions, and specifications for the Enterasys® B5 Gigabit Ethernet switches.

For information about the Command Line Interface (CLI) set of commands used to configure and manage the B5 switches, refer to the *Enterasys B5 CLI Reference*.

This preface provides an overview of this guide and the B5 manual set, and explains the symbols used throughout this guide.



Note: In this guide, the following terms are used:

- *Switch* refers to all B5 models, unless otherwise noted.
- *PoE* refers to Power over Ethernet (IEEE 802.3af and 802.3at).

Important Notice

Depending on the firmware version used in the switch, some features described in this document may not be supported. Refer to the Release Notes shipped with the B5 to determine which features are supported.

Who Should Use This Guide

This guide is intended for a network administrator responsible for installing and setting up the stackable switches.



Electrical Hazard: Only qualified personnel should perform installation procedures.

Riesgo Electrico: Solamente personal calificado debe realizar procedimientos de instalacion.

Elektrischer Gefahrenhinweis: Installationen sollten nur durch ausgebildetes und qualifiziertes Personal vorgenommen werden.

How to Use This Guide

Read through this guide completely to familiarize yourself with its contents and gain an understanding of the features and capabilities of the stackable Ethernet switch. A general knowledge of data communications networks is helpful when setting up the switch.

This guide provides information about the following:

For...	Refer to...
An overview of the B5 features.	Chapter 1, **Introduction**
Instructions to install the B5 on a flat surface or in a standard 19-inch rack, and configure the B5 in a stacked configuration. A description of Power over Ethernet (PoE) is also provided.	Chapter 2, **Installation**
Troubleshooting installation problems and diagnosing network/operational problems using the LEDs	Chapter 3, **Troubleshooting**
Specifications, environmental requirements, and physical properties of the B5 and optional pluggable transceivers	Appendix A, **Specifications**

Related Documents

The following manual can be obtained from the World Wide Web in Adobe Acrobat Portable Document Format (PDF) at the following site:

http://www.enterasys.com/support/manuals

Enterasys B5 CLI Reference describes how to use the Command Line Interface (CLI) to set up and manage the switches.

Conventions Used in This Guide

The following conventions are used in this guide:



Note: Calls the reader's attention to any item of information that may be of special importance.



Caution: Contains information essential to avoid damage to the equipment.

Precaución: Contiene información esencial para prevenir dañar el equipo.

Achtung: Verweißt auf wichtige Informationen zum Schutz gegen Beschädigungen.



Electrical Hazard: Warns against an action that could result in personal injury or death due to an electrical hazard.

Riesgo Electrico: Advierte contra una acción que pudiera resultar en lesión corporal o la muerte debido a un riesgo eléctrico.

Elektrischer Gefahrenhinweis: Warnung vor sämtlichen Handlungen, die zu Verletzung von Personen oder Todesfällen – hervorgerufen durch elektrische Spannung – führen können!



Warning: Warns against an action that could result in personal injury or death.

Advertencia: Advierte contra una acción que pudiera resultar en lesión corporal o la muerte.

Warnhinweis: Warnung vor Handlungen, die zu Verletzung von Personen oder gar Todesfällen führen können!

Getting Help

For additional support related to the product or this document, contact Enterasys Networks using one of the following methods:

World Wide Web	www.enterasys.com/services/support/
Phone	1-800-872-8440 (toll-free in U.S. and Canada) or 1-978-684-1888 For the Enterasys Networks Support toll-free number in your country: www.enterasys.com/services/support/contact/
Email	support@enterasys.com To expedite your message, please type **[B-Series]** in the subject line.
	To send comments or suggestions concerning this document to the Technical Publications Department: techpubs@enterasys.com To expedite your message, include the document part number in the email message.

Before contacting Enterasys Networks for technical support, have the following data ready:

- Your Enterasys Networks service contract number
- A description of the failure
- A description of any action(s) already taken to resolve the problem (for example, changing mode switches or rebooting the unit)
- The serial and revision numbers of all involved Enterasys Networks products in the network
- A description of your network environment (such as layout, cable type, other relevant environmental information)
- Network load and frame size at the time of trouble (if known)
- The device history (for example, if you have returned the device before, or if this is a recurring problem)
- Any previous Return Material Authorization (RMA) numbers

1

Introduction

This chapter introduces the Enterasys B5 Gigabit Ethernet stackable switches.

For information about...	Refer to page...
Overview	1-1
Features	1-5
PoE (Power over Ethernet) Support	1-5

Overview

The B5 Gigabit Ethernet switches are store and forward stackable switches that can be adapted and scaled to help meet your network requirements. These switches provide a management platform and an uplink to a network backbone for a stacked group of up to eight B5 switches.

You can also use a redundant DC power supply with these switches to help prevent downtime due to an internal power supply failure in the switch or AC power source.

You can install the switch on a flat surface or into a standard 19-inch rack with customer-supplied mounting hardware, and configure the switch functions using the WebView™ application, CLI switching commands, and/or SNMP.

The B5 switch can connect to Ethernet networks or workstations through RJ45 connectors, SFP ports, or 10G ports, depending on your model's configuration. The SFP and 10G ports support optional pluggable transceivers, which are hot swappable.

For pluggable transceiver compatibility and specifications, refer to the datasheet at http://www.enterasys.com/products/transceivers-ds.pdf.

Table 1-1 lists the number of active ports and port types for the B5 switches.

Table 1-1 B5 Switch Active Ports and Port Types

B5 Model	Ports	Number of Active Ports
B5G124-24 (see Figure 1-1 on page 1-2)	• 24 10/100/1000Base-T ports • Four combo SFP ports	26
B5G124-24P2 (see Figure 1-1 on page 1-2)	• 24 10/100/1000Base-T 802.3af and 802.3at PoE-capable ports • Four combo SFP ports	26
B5G124-48 (see Figure 1-2 on page 1-3)	• 48 10/100/1000Base-T ports • Four combo SFP ports	50
B5G124-48P2 (see Figure 1-2 on page 1-3)	• 48 10/100/1000Base-T 802.3af and 802.3at PoE-capable ports • Four combo SFP ports	50

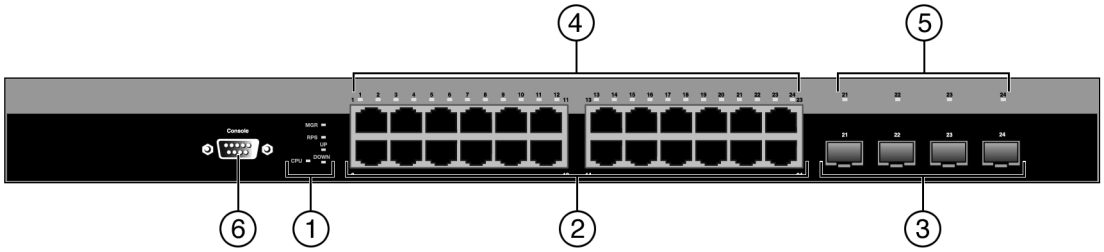
Table 1-1 B5 Switch Active Ports and Port Types (continued)

B5 Model	Ports	Number of Active Ports
B5K125-24 (see Figure 1-3 on page 1-3)	• 24 10/100/1000Base-T ports • Two combo SFP ports • Two 10G ports	26
B5K125-24P2 (see Figure 1-3 on page 1-3)	• 24 10/100/1000Base-T 802.3af and 802.3at PoE-capable ports • Two combo SFP ports • Two 10G ports	26
B5K125-48 (see Figure 1-4 on page 1-3)	• 48 10/100/1000Base-T ports • Two combo SFP ports • Two 10G ports	50
B5K125-48P2 (see Figure 1-4 on page 1-3)	• 48 10/100/1000Base-T 802.3af and 802.3at PoE-capable ports • Two combo SFP ports • Two 10G ports	50

Notes: Each combo SFP port supports the installation of 100BASE-FX, 1000BASE-SX, 1000BASE-LX, or 1000BASE-TX SFP pluggable transceivers.

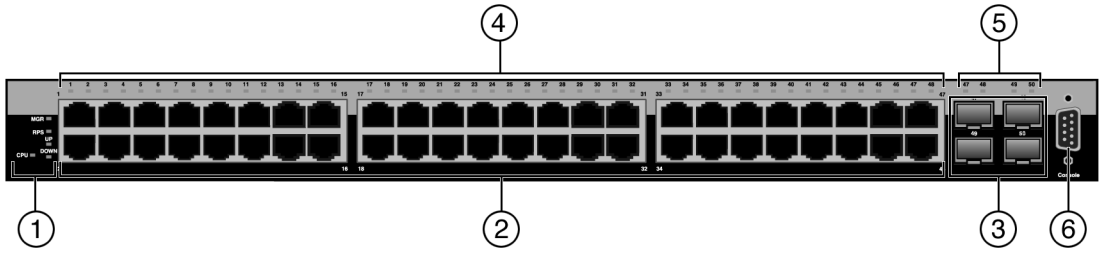
Each combo SFP port in use eliminates the availability of one RJ45 port. In other words, only 24 ports per slot can be active at any given time on components equipped with a combination of RJ45 and SFP interfaces. When an SFP transceiver in SFP port 23 establishes a link, RJ45 port 23 is disabled. When an SFP transceiver in SFP port 24 establishes a link, RJ45 port 24 is disabled.

Figure 1-1 B5G124-24 and B5G124-24P2 Front Panel



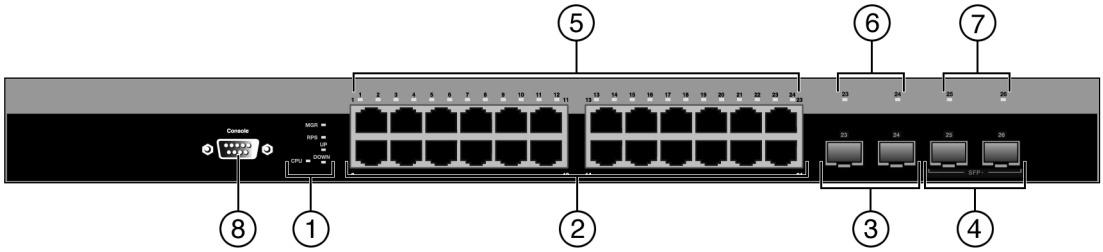
1 Switch status LEDs
2 RJ45 10/100/1000 Mbps ports
3 SFP slots
4 RJ45 port status LEDs
5 SFP port status LEDs
6 Console port

Figure 1-2 B5G124-48 and B5G124-48P2 Front Panel



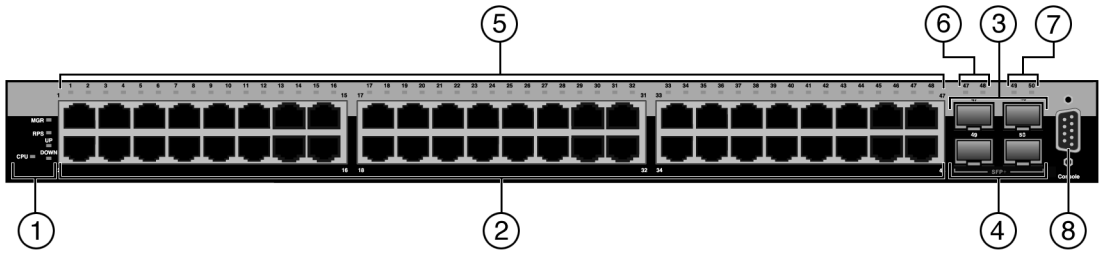
1	Switch status LEDs	**4**	RJ45 port status LEDs
2	RJ45 10/100/1000 Mbps ports	**5**	SFP port status LEDs
3	SFP slots	**6**	Console port

Figure 1-3 B5K125-24 and B5K125-24P2 Front Panel



1	Switch status LEDs	**5**	RJ45 port status LEDs
2	RJ45 10/100/1000 Mbps ports	**6**	SFP port status LEDs
3	SFP slots	**7**	10G port status LEDs
4	10G slots	**8**	Console port

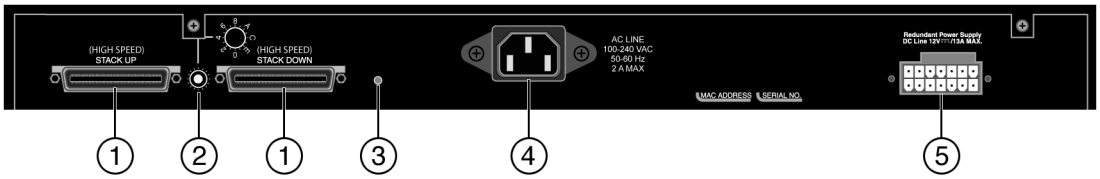
Figure 1-4 B5K125-48 and B5K125-48P2 Front Panel



1	Switch status LEDs	**5**	RJ45 port status LEDs
2	RJ45 10/100/1000 Mbps ports	**6**	SFP port status LEDs
3	SFP slots	**7**	10G port status LEDs
4	10G slots	**8**	Console port

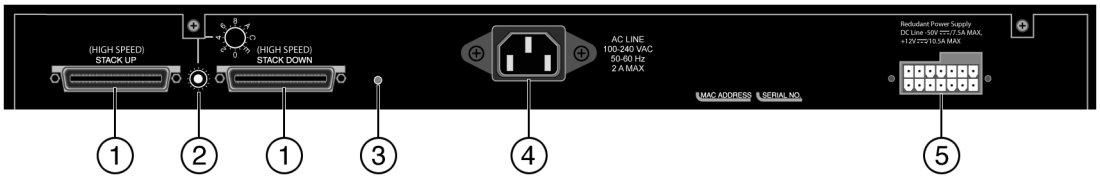
Figure 1-5 and Figure 1-6 show the back panels of the B5 switches.

Figure 1-5 B5 Switch Back Panels (Non-PoE B5 Switches)



1	Stack connection ports (shown without covers)	**4**	AC power input connector
2	Rotary switch	**5**	Redundant power supply (RPS) connector (shown without cover)
3	Password reset button		

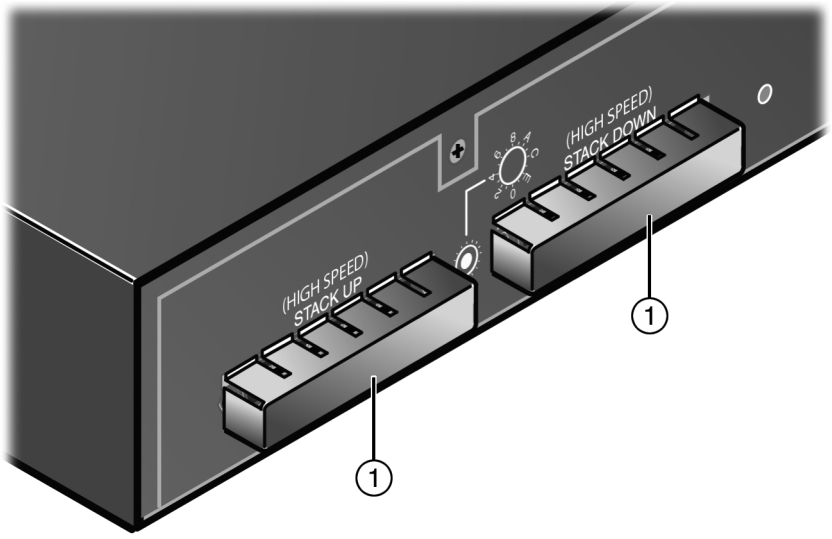
Figure 1-6 B5 Switch Back Panels (PoE B5 Switches)



1	Stack connection ports (shown without covers)	**4**	AC power input connector
2	Rotary switch	**5**	Redundant power supply (RPS) connector (shown without cover)
3	Password reset button		

Note: By default, the stack connection ports are covered with removable covers. If you are not connecting stacking cables, you must keep the covers on the stack connection ports to prevent EMI leakage. If you remove the stacking cables, you must replace the cover on the stack connection ports. See Figure 1-7.

Figure 1-7 Stack Connection Port Cover



1 Stack connection port covers

Features

High-Speed Stack Connections

The switches have rear-panel connectors for high-speed connections in a stack configuration. The high-speed stacking cables used for the connections are optional items and must be ordered separately. Stacking cables are available in four lengths: 30 centimeters, 1 meter, 2 meters, and 5 meters.

See "Connecting High-Speed Stacking Cables" on page 2-6 for details on these cables.



Note: Use only the black high-speed stacking cables with B5 switches. Do not use the stacking cables that were designed for use with other B-Series devices.

Redundant Power Supply Capability

The B5 has power supply redundancy capability when connected to an optional external redundant power supply:

- STK-RPS-150PS, a 102–156 watt DC power supply for redundant power for non-PoE B5 switches.
- STK-RPS-500PS, a 500 watt DC power supply for redundant power for PoE B5 switches.

If the internal power supply fails, the RPS assumes the entire load of the switch without interrupting network traffic. The internal power supply and RPS each have their own AC power connection, which enables the connection of each power supply to a different AC power circuit for additional AC power source redundancy.

Rotary Switch

Currently, the rotary switch is for future use only.

PoE (Power over Ethernet) Support

The B5 PoE-capable models (B5G124-24P2, B5G124-48P2, B5K125-24P2, and B5K125-48P2) are IEEE 802.3af and 802.3at compliant, which means they can provide power over Ethernet (PoE) cable connections from their RJ45 connectors to PDs in the network.

PoE, defined in IEEE standards 802.3af and 802.3at, refers to the ability to provide 54 Vdc (for 802.3at) or 48 Vdc (for 802.3af) operational power through an Ethernet cable from a switch or other device that can provide a PoE-compliant port connection to a powered device (PD). Examples of PDs include:

- Voice over IP devices such as PoE-compliant digital telephones
- Devices that support Wireless Application Protocol (WAP) such as wireless access points
- Security cameras
- Virtual desktops

The B5 PoE-capable models support the standard resistor-based detection method, as well as AC disconnect capability.

Table 1-2 lists, for each B5 PoE-compliant model, the number of ports that can provide 802.3af and 802.3at PoE power.

Table 1-2 Number of B5 Ports Providing PoE Power Simultaneously

	Ports Providing PoE Power Simultaneously	
B5 Model	**802.3af**	**802.3at (30W)**
B5G124-24P2	24 ports (15.4W)	12 ports
B5K125-24P2	24 ports (15.4W)	12 ports
B5G124-48P2	24 ports (15.4W), 48 ports (7.5W)	12 ports
B5K125-48P2	24 ports (15.4W), 48 ports (7.5W)	12 ports

2

Installation



Electrical Hazard: Only qualified personnel should perform installation procedures.

Riesgo Electrico: Solamente personal calificado debe realizar procedimientos de instalacion.

Elektrischer Gefahrenhinweis: Installationen sollten nur durch ausgebildetes und qualifiziertes Personal vorgenommen werden.

Important Notice

Read the Release Notes shipped with the switch to check for any exceptions to the supported features and operation documented in this guide.

This chapter provides instructions to install the B5. Unless otherwise noted, the instructions apply to all B5 models. Follow the order of the sections listed below to correctly install the switch.

For information about...	Refer to page...
Considerations Prior to Installation	2-1
Required Tools	2-2
Unpacking the Switch	2-2
Installing the Switch on a Flat Surface	2-2
Connecting High-Speed Stacking Cables	2-6
Configuring Switches in a Stack	2-8
Connecting AC Power	2-10
Installing and Connecting B5 Redundant Power Systems	2-11
Connecting to the Console Port for Local Management	2-22
Connecting to the Network	2-25
Completing the Installation	2-30

Considerations Prior to Installation

When installing the switch, note the following:

- Before starting the installation procedure, notify the network administrator of the installation.
- Follow the installation procedures in the order as presented in this guide.
- Do not connect the switch to the network until you have established the correct IP address.

Required Tools

A Phillips screwdriver is required to install the switch into a rack.

Unpacking the Switch

Unpack the switch as follows:

1. Open the box and remove the packing material protecting the switch.
2. Verify that the contents of the carton contains the items listed in the table below.

Item	Quantity
B5 switch	1
Mounting kit, consisting of two mounting brackets and eight M3x6-mm screws for rack mounting	1
Rubber feet with adhesive backing for installation on a flat surface	4
AC power cord (The type of power cord is country dependent.)	1
DB9 female-to-DB9 female Console Cable	1
Quick Start Guide	1

3. Remove the tape seal on the non-conductive bag to remove the switch.
4. Perform a visual inspection of the switch for any signs of physical damage. Contact Enterasys Networks if there are any signs of damage. Refer to "Getting Help" on page xvii for details.

Installing the Switch on a Flat Surface

When installing the switch on a flat surface, the installation of the rubber feet is recommended to prevent the switch from sliding on a flat surface. For instructions to rack mount the switch, proceed to "Rack Mounting the Switch" on page 2-4.

Installing the Rubber Feet

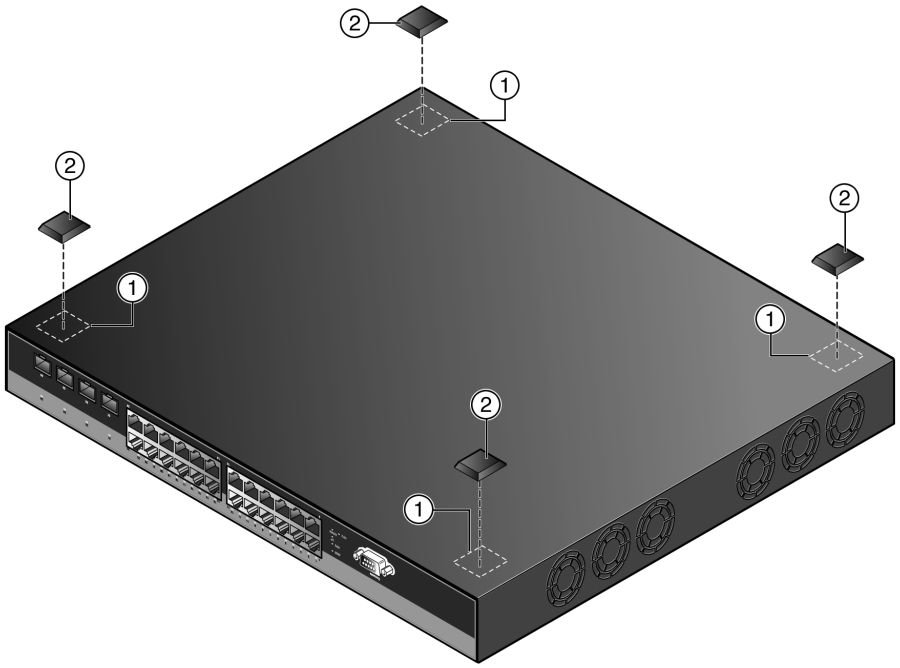


Note: Do not install the rubber feet if you are going to rack mount the switch.

To install the rubber feet, refer to Figure 2-1 and proceed as follows:

1. Place the switch on its back on a sturdy flat surface to gain access to the bottom of the chassis.
2. Remove the four rubber feet from their plastic bag in the shipping box.

Figure 2-1 Chassis Bottom, Rubber Feet Placement



1 Locations to install the rubber feet (four locations) **2** Rubber feet with adhesive backing (four)

3. Remove the protective strip from the back of one rubber foot and position it on a corner and press firmly into place.

   Repeat this procedure to install the remaining three rubber feet in the other three corners.

4. After installing the rubber feet, return the switch to its upright position.

Note: If a number of switches are being installed in a stack, repeat steps 1 through 4 to install the rubber feet on each switch before continuing with the installation.

5. Proceed to "Guidelines for Flat Surface Installation" on page 2-3. For a rackmount installation, proceed to "Rack Mounting the Switch" on page 2-4.

Guidelines for Flat Surface Installation

Locate the switch within 182.88 cm (6 ft) of its power source and on a surface as shown in Figure 2-2. If an optional redundant power system is going to be installed and connected to the 14-pin Redundant Power Supply input connector on the rear of the switch, refer to the installation guide shipped with the redundant power system.

Caution: To ensure proper ventilation and prevent overheating, leave a minimum clearance space of 5.1 cm (2.0 in.) at the left and right of the switch.

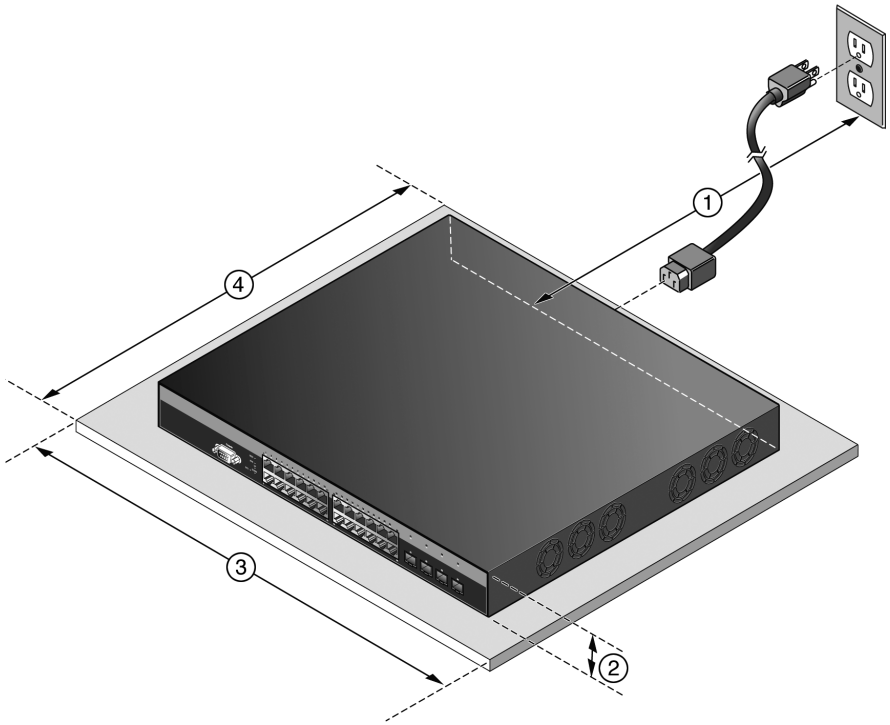
Do not connect the switch to the AC power source until instructed to do so later in the installation process.

Precaución: Para asegurar una buena ventilación y evitar que el sistema se sobrecaliente, deje un espacio mínimo de 5.1 cm (2 pulgadas) con respecto a los lados y a la parte posterior del aparato.

No conecte el dipositivo a la fuente primaria hasta que no se le indique.

If you are installing several switches in a stack, proceed to "Connecting High-Speed Stacking Cables" on page 2-6. If the switch is being installed as a standalone switch, proceed to "Connecting AC Power" on page 2-10 for power connection instructions.

Figure 2-2 Area Guidelines for Switch Installation on Flat Surface



1	Approximately 152 cm (5 ft) from power source	**3**	44.5 cm (19.4 in.) for proper ventilation
2	4.45 cm (1.75 in.) per switch. (Vertical clearance depends on number of switches stacked.)	**4**	41.9 cm (16.5 in.) for proper ventilation

Rack Mounting the Switch

To install the switch in a 19-inch rack, you need:

- Two rackmount brackets and mounting screws (rackmount kit) shipped with the switch.
- Four customer-supplied screws to attach the switch to a standard 19-inch rack.

Guidelines for Rackmount Installation

The installation site must be within reach of the network cabling and meet the requirements listed below:

- Appropriate grounded power receptacles must be located within 152 cm (5 ft) of the location.
- A temperature of between 0°C (32°F) and 50°C (122°F) must be maintained at the installation site with fluctuations of less than 10°C (18°F) per hour.

Note: To ensure proper ventilation and prevent overheating, leave a minimum clearance space of 5.1 cm (2.0 in.) at the left and right of the switch.

Warning: Before rack-mounting the switch, ensure that the rack can support it without compromising stability. Otherwise, personal injury and/or equipment damage may result.

Advertencia: Antes de montar el equipo en el rack, asegurarse que el rack puede soportar su peso sin comprometer su propia estabilidad, de otra forma, daño personal o del equipo puede ocurrir.

Warnhinweis: Überzeugen Sie sich vor dem Einbau des Gerätes in das Rack von dessen Stabilität, ansonsten könnten Personenschäden oder Schäden am Gerät die Folge sein.

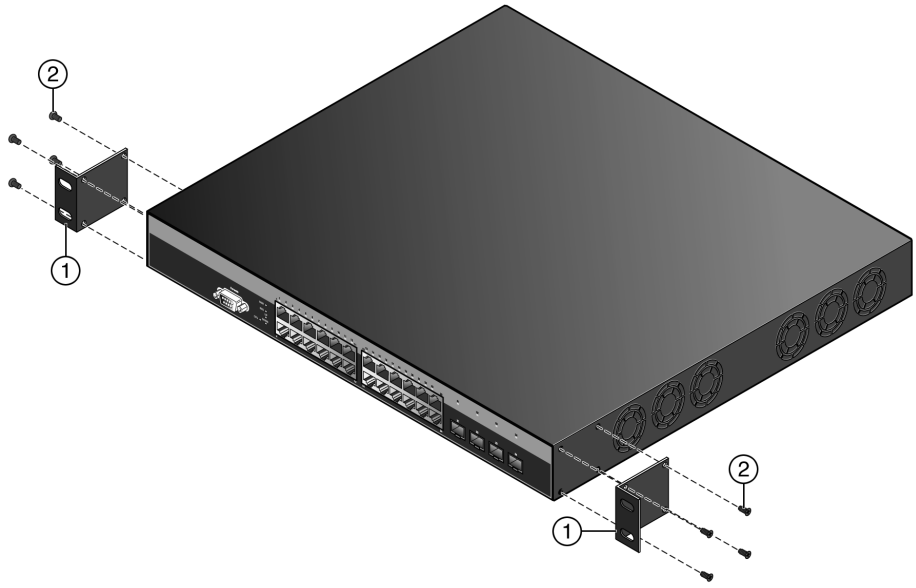
Note: Do not install the rubber feet if you are rack mounting the switch.

Attaching the Brackets and Installing in a Rack

Proceed as follows to install the switch into a 19-inch rack:

1. Attach the rackmount brackets to the switch, as shown in Figure 2-3, using the eight M3x6 mm flathead screws shipped with the switch.

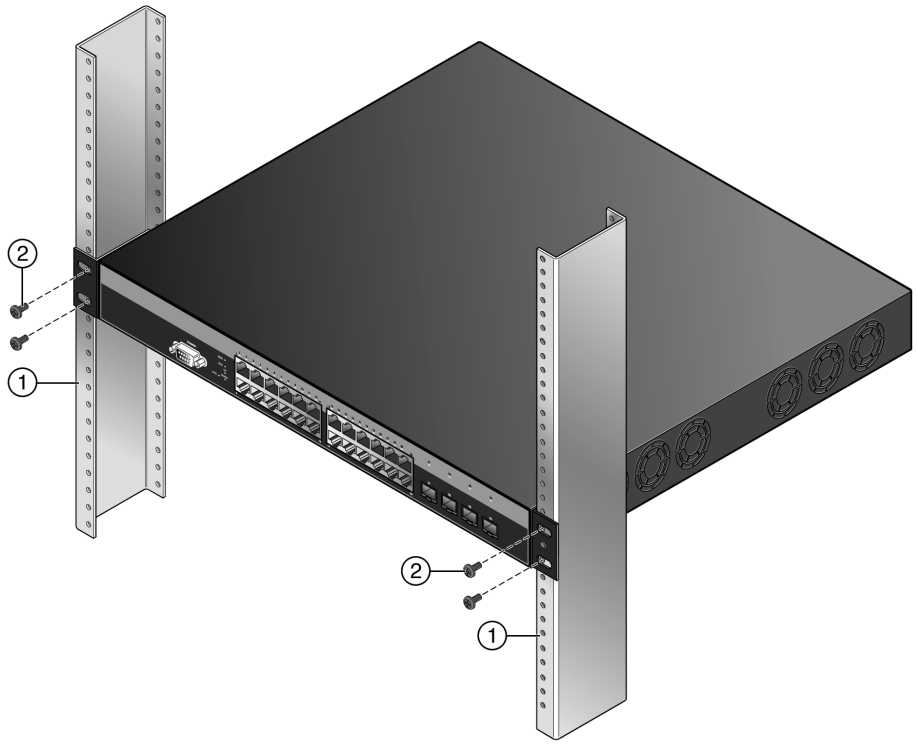
Figure 2-3 Attaching the Rackmount Brackets



1 Rackmount brackets **2** M3x6 mm flathead screws

2. With the mounting brackets attached, position the switch between the vertical frame members of the 19-inch rack as shown in Figure 2-4. Then fasten the switch securely to the frame using four customer-supplied mounting screws.

Figure 2-4 Fastening the Switch to the Rack



1 Rails of 19-inch rack **2** Mounting screws

3. If you are installing this switch in a stacked configuration, repeat this procedure for each switch until all switches have been installed in the stack, then proceed to "Connecting High-Speed Stacking Cables" on page 2-6. Otherwise, proceed to "Connecting AC Power" on page 2-10.

Connecting High-Speed Stacking Cables

Note: By default, the stack connection ports are covered with removable covers. If you are not connecting stacking cables, keep the covers on the stack connection ports. If you remove the stacking cables, you must replace the covers on the stack connection ports.

The stack of switches can be connected in a ring or daisy chain topology. In a ring topology all the switches are connected. The only difference in the cable connections in a daisy chain topology is that one cable is not installed. Up to eight switches can be stacked together and connected by high-speed stacking cables. You can add switches and reach up to a maximum of 384 Ethernet ports per stack. The high-speed stacking cables allow the entire stack to operate with a single IP address.

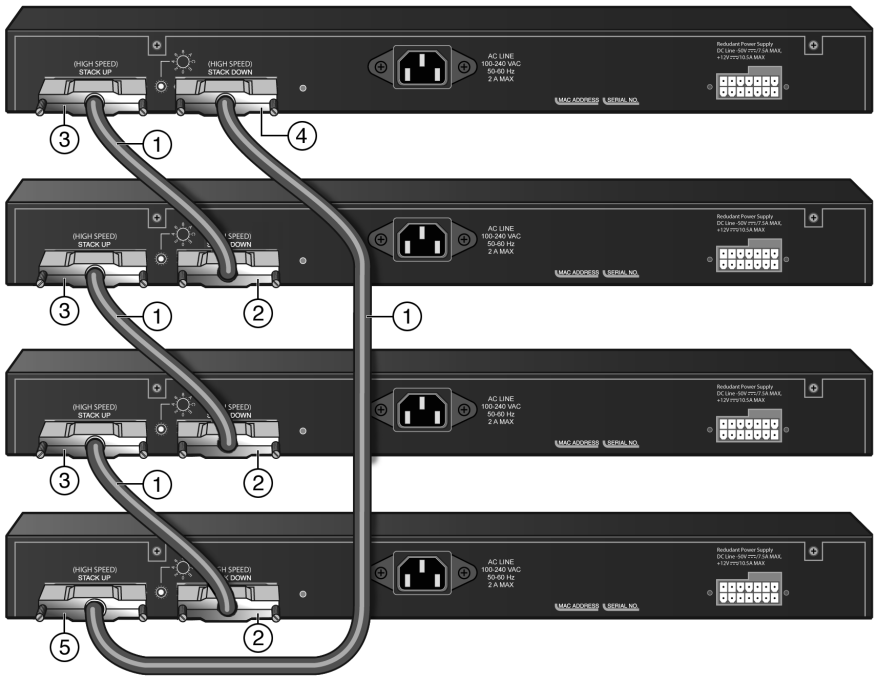
The high-speed stacking cables are optional items that you must order separately. The B5 supports the following stacking cables:

- STK-CAB-SHORT, a 30cm cable
- STK-CAB-LONG, a 1m cable
- STK-CAB-2M, a 2m cable
- STK-CAB-5M, a 5m cable

Figure 2-5 shows an example of a four-high stack connected in a ring topology. All STACK DOWN and STACK UP connectors are used in the installation. The high-speed stacking cable connections

are from the STACK DOWN connector of one switch to the STACK UP connector of the next switch up in the stack. A high-speed stacking cable connection from the STACK DOWN connector of the switch at the top of the stack to the STACK UP connector at the bottom of the stack completes the ring connection. In a daisy chain topology, one cable connection is not made.

Figure 2-5 High-Speed Stacking Cable Connections



1 High-speed stacking cable
2 STACK DOWN connector
3 STACK UP connector
4 STACK DOWN connector at top of stack
5 STACK UP connector at bottom of stack

Note: Hand tighten the screws on the stacking cables to ensure good connections to the STACK DOWN and STACK UP connectors.

After connecting the cables to the stacking ports, proceed to "Configuring Switches in a Stack" on page 2-8 for instructions.

Configuring Switches in a Stack

The information in the following sections is important to understand B5 switch operation and installations in a stack installation.

About B5 Switch Operation in a Stack

The B5 switches are stackable switches that can be adapted and scaled to help meet your network needs. These switches provide a management platform and uplink to a network backbone for a stacked group of up to eight B5 switches.



Note: You can stack a B5 switch only with other B5 switches. You cannot stack a B5 switch with swtiches that are not B5 switches.

Once installed in a stack, the switches behave and perform as a single switch. As such, you can start with a single switch and add more switches as your network expands. You can also mix different products in the family in a single stack to provide a desired combination of port types and functions to match the requirements of individual applications. In all cases, a stack of switches performs as one large product, and is managed as a single network entity.

When switches are installed and connected as described in "Connecting High-Speed Stacking Cables" on page 2-6, the following occurs during initialization:

- The switch that will manage the stack is automatically established and is referred to as the manager switch.
- All other switches are established as member switches in the stack.
- The hierarchy of the switches that will assume the function of backup manager is also determined in case the current manager malfunctions, is powered down, or is disconnected from the stack.
- The Console port on the manager switch remains active for out-of-band (local) switch management, but the Console port on each member switch is deactivated. This enables you to set the IP address and system password using a single Console port. Now each switch can be configured locally using only the manager's Console port, or in-band using a remote device and the CLI set of commands described in this section. For procedures used for various types of connections to the Console port, refer to "Connecting to the Console Port for Local Management" on page 2-22.

Once a stack is created (more than one switch is interconnected), the following occurs:

1. Switch (unit) IDs are arbitrarily assigned on a first-come, first-served basis.
2. Switch IDs are saved against each module. Then, every time a switch is power-cycled, it will initialize with the same switch ID. This is important for port-specific information (for example: ge.4.12 is the 12th Gigabit Ethernet port on switch number 4).
3. The management election process uses the following precedence to assign a manager switch:
    a. Previously assigned/elected maanger switch
    b. Management assigned priority (values 1–15)
    c. Hardware preference level
    d. Highest MAC Address

Stack Manager Selection

When you install and connect all the high-speed stacking cables to the switches in the stack, the following occurs once power is applied to the switches:

- The switch that will manage (manager) the stack is automatically established with all other switches established as member switches in the stack. When the switches complete their initializations, one of the switches in the stack will illuminate its MGR LED, indicating that it is the stack manager. The MGR LED on each member switch will be off.
- The hierarchy of the switches that will function as backup manager is also determined in case the current manager malfunctions, is powered down, or is disconnected from the stack.
- The Console port on each member switch is deactivated. Only the Console port on the manager switch is active for out-of band configuration to set the IP address, password, and other configuration settings.

Once you know which switch is the manager, proceed to "Connecting to the Console Port for Local Management" on page 2-22.

Recommended Procedures for New and Existing Stacks

Important

The following procedures assume that all switches have a clean configuration from manufacturing. When adding a new switch to an already running stack, it is also assumed that the new switch is using the same firmware image version as other switches in the stack.

Installing a New Stackable System of Up to Eight Switches

Use the following procedure to install a new stack of up to eight switches out of the box. Before applying power, make **all** physical connections with the stack cables as described in "Connecting High-Speed Stacking Cables" on page 2-6.

1. Once all of the stack cables have been connected, individually power on each switch from top to bottom (connecting power to a switch is described in "Connecting AC Power" on page 2-10).

   If the switches are powered on almost simultaneously, the system will automatically select the first one that powers up as the Master switch and the others as member switches. The switches are assigned unit IDs in the order that they become fully operational.

   You can control the unit ID assignment according to the physical position in a stack. When you power up each switch and allow it to become fully operational before applying power to the next switch, the first one becomes the manager and all the next switches will join that stack (regardless of priority, firmware revision, or MAC address). The switches are assigned unit IDs in the order that you power on each switch.

   Note: Once switch IDs are assigned, they are persistent and will be retained during a power cycle to any or all of the switches.

2. Connect to Console port of the manager switch.
3. (Optional) If desired, change the management switch using the **set switch movemanagement** command as described in the *Enterasys B5 CLI Reference*.
4. Once the desired master switch has been selected, reset the system using the **reset** command as described in the *Enterasys B5 CLI Reference*.

Adding a New Switch to an Existing Stack

Use the following procedure to install a new switch to an existing stack configuration.

Note: This procedure assumes that the new switch being added has a clean configuration from manufacturing and is running the same firmware image version as other switches in the stack. The new switch being added also must have the same license as the manager switch.

1. Ensure that power is off on the new switch being installed.
2. Use one of the following methods to complete the stack cable connections:
   - If the running stack uses a daisy chain topology, make the stack cable connections from the bottom of the stack to the new switch (that is, STACK DOWN connector from the bottom switch of the running stack to the STACK UP connector on the new switch).
   - If the running stack uses a ring stack topology, break the ring and make the stack cable connections to the new switch to close the ring.
3. Apply power to the new switch.

Important Considerations About Using Clear Config in a Stack

When using the **clear config** command (as described in the *Enterasys B5 CLI Reference*) to clear configuration parameters in a stack, it is important to remember the following:

- Use **clear config** to clear config parameters without clearing stack switch IDs. This command WILL NOT clear stack parameters and avoids the process of re-numbering the stack.
- Use **clear config all** when it is necessary to clear all config parameters, including stack switch IDs and switch priority values.

Connecting AC Power

Note: The power supply in the switch has automatic voltage sensing that allows connection to power sources ranging from 100 to 240 VAC.

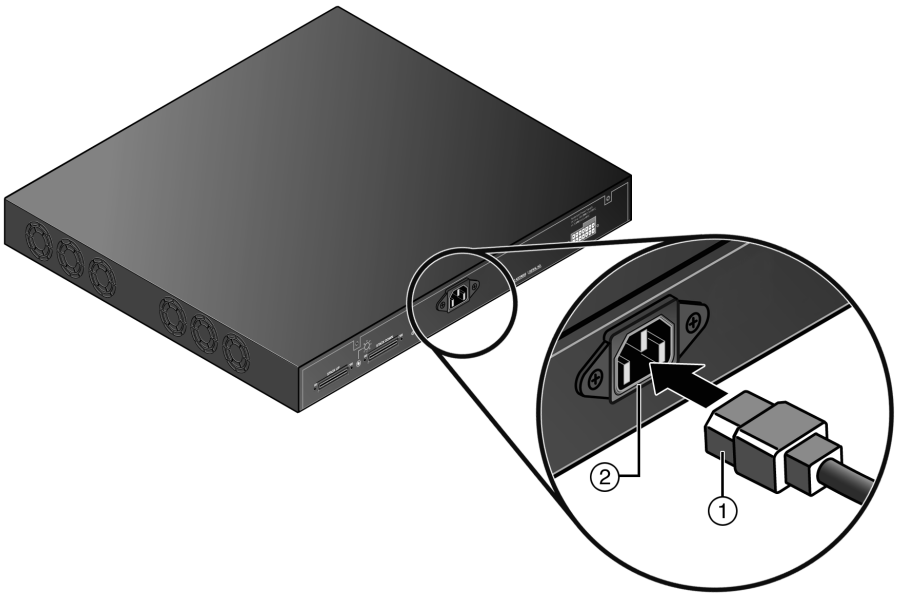
To connect a switch to the AC power source, refer to Figure 2-6 and proceed as follows:

1. Plug the power cord into the switch AC power connector.
2. Plug the other end of the power cord into a dedicated grounded AC outlet (not shown). The type of power outlet and power cord are country-dependent.

Note: If you plan to connect all the switches quickly to allow automatic Manager selection, wait until all switches are fully operational before proceeding to the next step.

If you plan to power up each switch and allow it to become fully operational before applying power to the next switch, proceed to the next step.

Figure 2-6 Switch Rear View



1 AC power cord **2** AC power connector

3. Observe that the power CPU LED (not shown), located on the front panel. During the initialization, the CPU LED will start by illuminating solid amber, then start blinking green, then blinking amber, then blinking green again until the end of the initialization, and then turns solid green.

Note: If the CPU LED illuminates solid red, there was a critical failure. For more information about the LED indications and troubleshooting, refer to Chapter 3, **Troubleshooting**. If you need additional help, contact Enterasys Networks. Refer to "Getting Help" on page xvii for details.

If the switch is a standalone switch, it will take approximately 30 seconds for the switch to start up. If the switch is a stack manager, it can take up to 3 minutes or more to start up, depending on the number of member switches in the stack.

Installing and Connecting B5 Redundant Power Systems

Depending on your B5 switch, you can install one of the following redundant power supplies:

- STK-RPS-150PS, a 150W redundant power supply for non-PoE B5 models.
- STK-RPS-500PS, a 500W redundant power supply for PoE-compliant B5 models.

STK-RPS-150PS

You can install an STK-RPS-150PS in the following RPS shelves:

- STK-RPS-150CH2, a two-slot chassis
- STK-RPS-150CH8, an eight-slot chassis

Required Tools

A flat-blade screwdriver is required to install the STK-RPS-150CH2 or STK-RPS-150CH8 shelf and STK-RPS-150PS power supplies.

Unpacking the Shelf and Power Supply

The shelf and the power supply are shipped separately. To unpack these devices proceed as follows:

1. Open the box and remove the packing material protecting the shelf or power supply.
2. Verify the contents of each carton and compare the contents shipped with those listed in Table 2-1, Table 2-2, and Table 2-3.
3. Perform a visual inspection of the switch devices for any signs of physical damage. Contact Enterasys Networks if there are any signs of damage. Refer to "Getting Help" on page xvii for details.

Table 2-1 Contents of STK-RPS-150CH2 Carton

Item	Quantity
STK-RPS-150CH2	1
Quick Reference	1
Release Notes	1

Table 2-2 Contents of STK-RPS-150CH8 Carton

Item	Quantity
STK-RPS-150CH8	1
Quick Reference	1
Release Notes	1

Table 2-3 Contents of STK-RPS-150PS Carton

Item	Quantity
STK-RPS-150PS 150 watt DC power supply	1
USA, NEMA power cord	1
RPS cable	1
Notice card	1

Installing an STK-RPS-150PS



Caution: Observe all Electrostatic Discharge (ESD) precautions when handling sensitive electronic equipment.

Precaución: Al trabajar con equipos electrónicos sensibles, tome todas las precauciones de seguridad para evitar descargas de electricidad estática.

When you receive your STK-RPS-150CH2 or STK-RPS-150CH8 shelf, a coverplate will be in place over each power-supply slot.



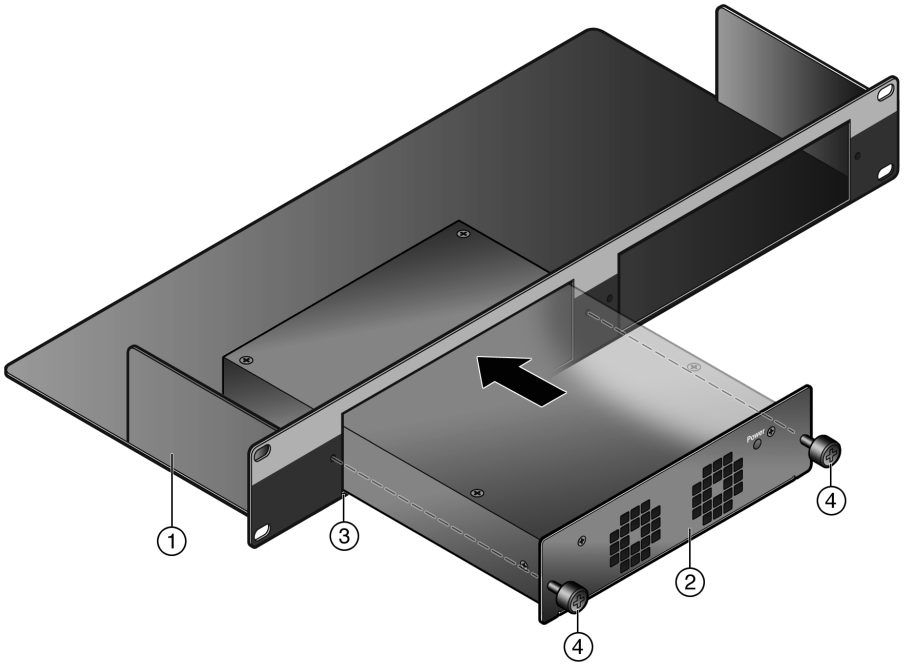
Note: To prevent EMI leakage from the shelf, keep coverplates installed over any empty power supply slots.

To install the power supplies, refer to Figure 2-7 or Figure 2-8 and proceed as follows:

1. Place the chassis on a sturdy flat surface where you plan to install power supplies.
2. Align the power supply with one of the slots, then slide the power supply forward until its front panel is flush against the chassis front panel.

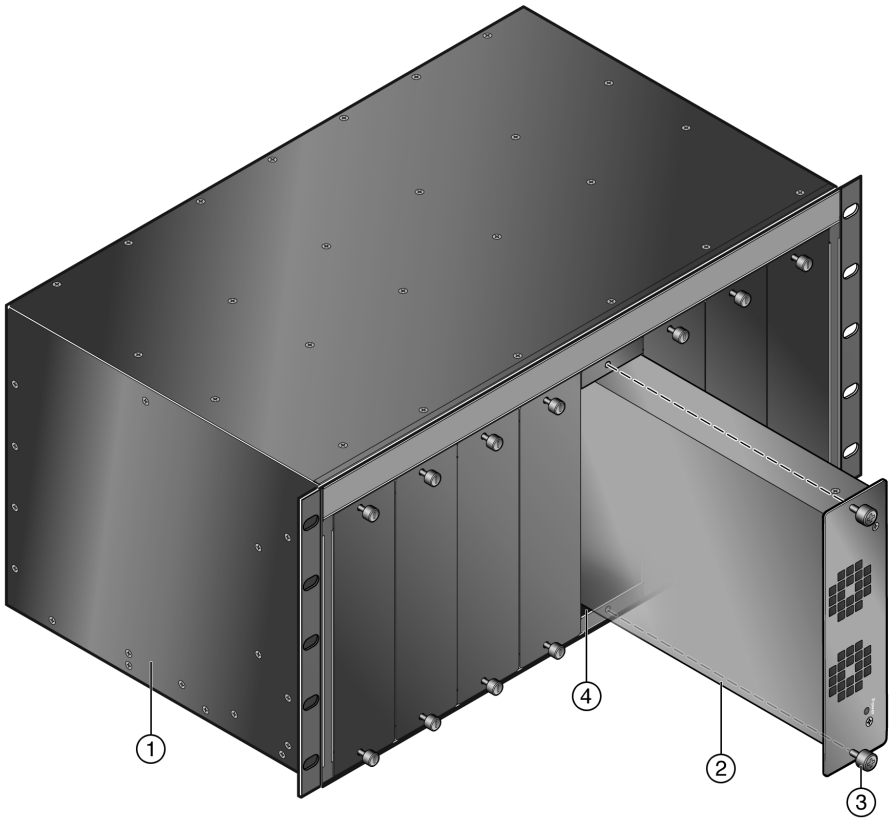
3. Fasten the power supply to the chassis using the captive screws on the PSM front panel.
4. Repeat steps 2 and 3 for each additional power supply.
5. Proceed to "Installing the Shelf into the Rack" on page 2-15 for the rack mount installation instructions.

Figure 2-7 STK-RPS-150PS Installation in an STK-RPS-150CH2 Shelf



1 STK-RPS-150CH2 shelf
2 STK-RPS-150PS power supply
3 Shelf power supply slot
4 Captive screws (2)

Figure 2-8 STK-RPS-150PS Installation in an STK-RPS-150CH8 Shelf



1	STK-RPS-150CH8 shelf	**3**	Captive screws (2)
2	STK-RPS-150PS power supply	**4**	Chassis power supply slot

Removing an Installed STK-RPS-150PS

To remove a power supply installed in an operating system, proceed as follows:

Caution: Observe all Electrostatic Discharge (ESD) precautions when handling sensitive electronic equipment.

Precaución: Al trabajar con equipos electrónicos sensibles, tome todas las precauciones de seguridad para evitar descargas de electricidad estática.

1. Unplug the AC power cord of the power supply from the AC power source first, then from the rear of the power supply .
2. Loosen the captive screws securing the power supply until it is released from the chassis front panel.
3. Pull the power supply out and remove it from the chassis.
4. Repeat steps 1 through 3 for each additional power supply you plan to remove.

Note: If you plan to remove and not replace an STK-RPS-150PS, immediately reinstall coverplates over any empty power supply slots to prevent EMI leakage from the chassis.

Installing the Shelf into the Rack

To install an STK-RPS-150CH2 or STK-RPS-150CH8 into a 19-inch (48.3-cm) rack, you need the following:

- For the STK-RPS-150CH2: four customer-supplied screws to fasten the shelf to the rack rails
- For the STK-RPS-150CH8: ten customer-supplied screws to fasten the shelf to the rack rails

After installing the power supplies as described back in "Installing an STK-RPS-150PS" on page 2-12, rack mount the chassis as follows:

1. Refer to the installation guidelines ("Guidelines for Rackmount Installation" on page 2-15).
2. Install the chassis into the rack ("Rack Mounting the Chassis" on page 2-16).
3. Connect the RPS cables and AC power cords of the power supplies ("Connecting the RPS Cable and AC Power Cord" on page 2-17).

Guidelines for Rackmount Installation

The installation site must be within reach of the network cabling and meet the requirements listed below:

- Shelf placement musst be close enough to connect the 2-meter (6.6-foot) AC power cords from the power supplies to the AC power source.
- Up to two three-pronged power receptacles capable of delivering the current and voltage specified in "STK-RPS-150PS Specifications" on page A-5. Up to two AC outlets on independently-fused circuits must be within 182 centimeters (6 feet) from the installation site. If there is an AC power source failure, this will prevent the powering down of PSMs due to a single source power failure.
- Power cords and type of outlet are dependent on the country. In the United States, one power cord with an NEMA 5-15P plug is provided with each STK-RPS-150PS.
- An ambient temperature of between 5°C (41°F) and 40°C (104°F) must be maintained at the installation site with fluctuations of less than 10°C (18°F) per hour.



Caution: To ensure proper ventilation and prevent overheating, leave a minimum clearance space of 10.16 cm (4.0 in.) at the front and rear of the B5 chassis.

Precaución: Para asegurar una buena ventilación y evitar que el sistema se sobrecaliente, deje un espacio mínimo de 10.16 cm (4 pulgadas.) con respecto a la parte delantera y trasera del chasis B5.



Warning: Before installing the chassis into a rack, ensure that the rack can support the device(s) without compromising the stability of the rack. Otherwise, personal injury and/or equipment damage may result.

Advertencia: Antes de instalar el chassis en un rack, asegurarse que el rack puede soportar el(los) dispositivo(s) sin comprometer la estabilidad del mismo. De otra forma puede suceder algun tipo de daño personal o del equipo.

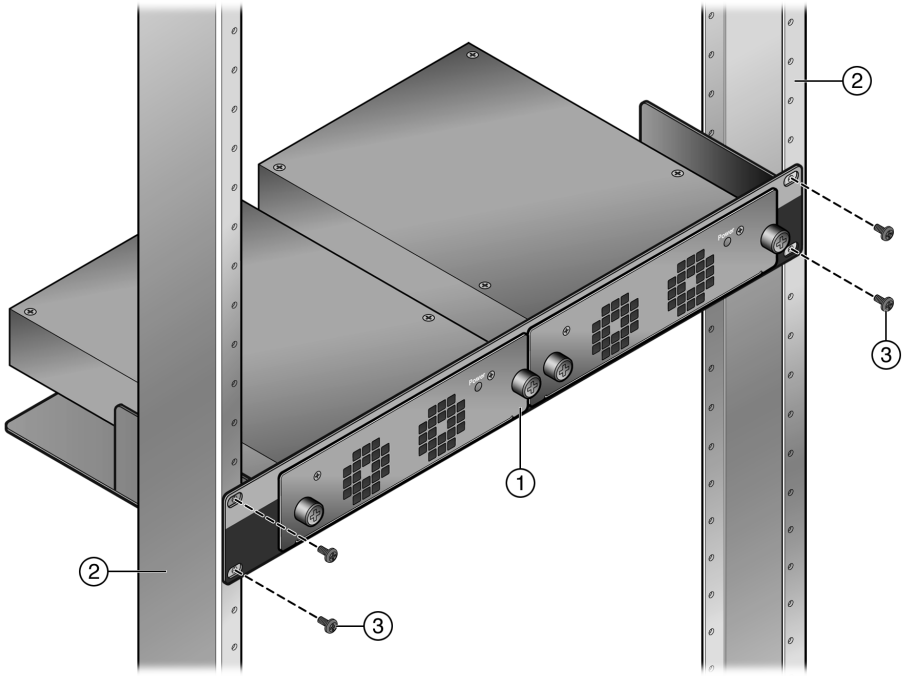
Warnhinweis: Schützen Sie sich vor Verletzungen und Geräteschaden, überzeugen Sie sich vor der Installation des Chassis in das Rack, von dessen Stabilität.

Rack Mounting the Chassis

Refer to Figure 2-9 or Figure 2-10 and proceed as follows to install the chassis into a 19-inch (48.3-cm) rack:

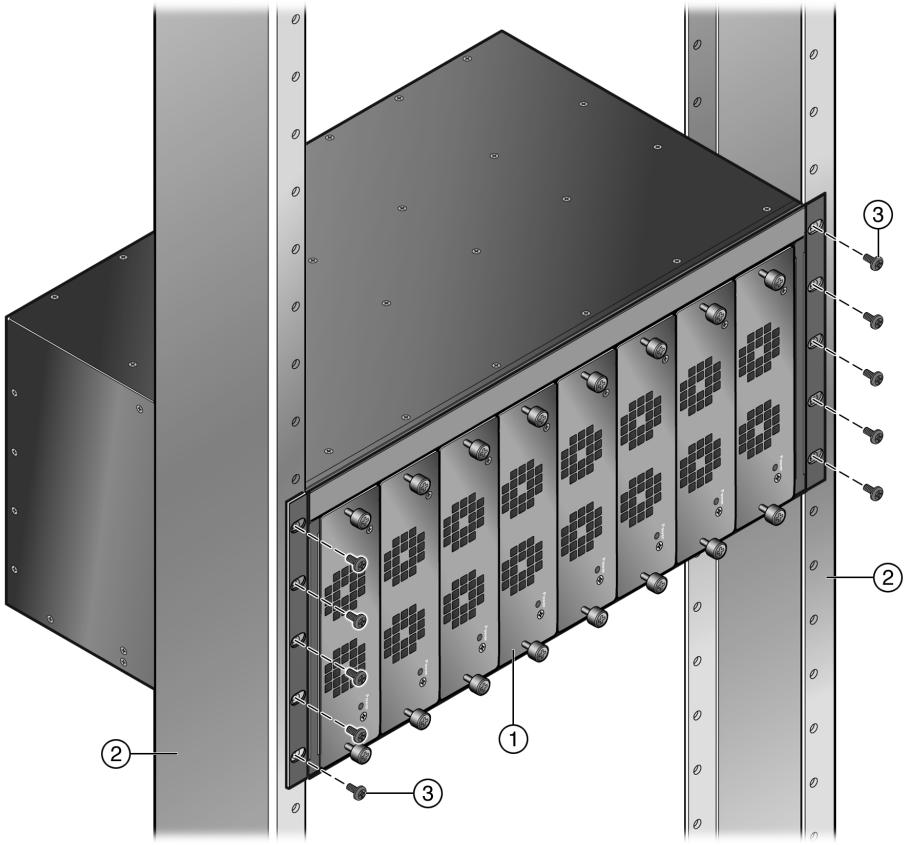
1. Position the chassis between the vertical rails and align the mounting holes in the chassis brackets with those in the rack frame.
2. Fasten the chassis securely to the rails using the customer-supplied mounting screws (four for the STK-RPS-150CH2, ten for the STK-RPS-150CH8).

Figure 2-9 Fastening the STK-RPS-150CH2 to the Rack



1 STK-RPS-150CH2 chassis
2 Rails of 19-inch rack
3 Mounting screws

Figure 2-10 Fastening the STK-RPS-150CH8 to the Rack



1 STK-RPS-150CH8 chassis
2 Rails of 19-inch rack
3 Mounting screws

Connecting the RPS Cable and AC Power Cord

The redundant power supply is connected to a B5 Ethernet switch switch using an RPS cable.

To connect a redundant power supply, proceed as follows:

1. Using a Philips screwdriver, remove the cover from the redundant power supply connector on the B5 switch.
2. Connect one end of the RPS cable to the redundant power supply connector on the B5 switch. Then connect the other end of the cable to the redundant power supply connector at the rear of the RPS as shown in Figure 2-11.
3. Connect the AC power cord to the AC input power connector on the RPS shown in Figure 2-12, then plug the AC power cord into the main AC power outlet.

   The green Power LED on the front of the RPS will illuminate to indicate a successful connection. If the LED remains off, proceed as follows:

   a. Check the AC power cord connection at the AC power source and make sure the power source is within specification.

   b. Check the AC power connection to the RPS.

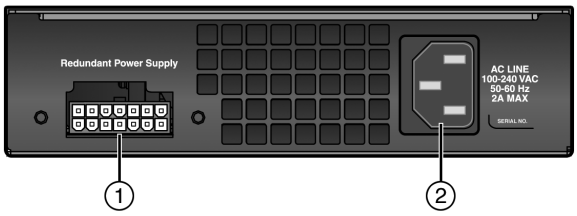
   c. Swap the AC power cord with a known good one.

   d. If the green LED continues to remain off, contact Enterasys Networks. Refer to "Getting Help" on page xvii for instructions. Otherwise, proceed to step 4.

On certain switches, an LED indicator will show that a redundant power supply is now in operation.

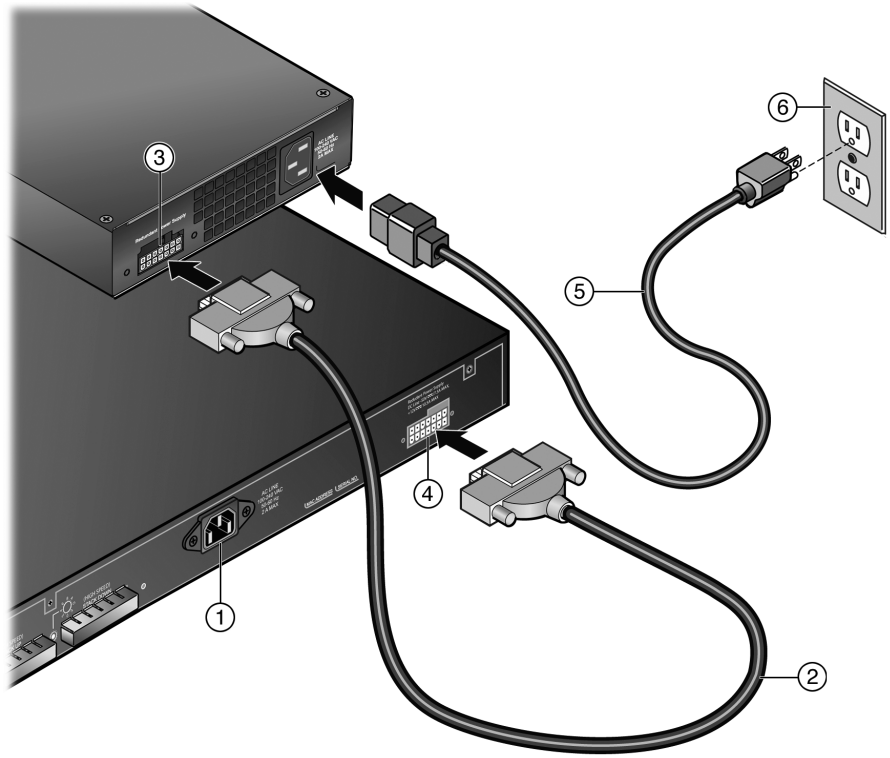
Note: No change in switch configuration is necessary for this installation.

Figure 2-11 Power Connectors on STK-RPS-150PS (rear view)



1 Redundant power supply connector
2 AC power connector

Figure 2-12 STK-RPS-150PS RPS Cable and AC Power Cord Connections



1 B5 switch
2 RPS cable
3 STK-RPS-150PS redundant power supply connector
4 B5 switch redundant power supply connector
5 AC power cord (type varies depending on country)
6 AC power outlet with ground connection (type varies depending on country)

4. On certain switches, an LED indicator will show that a redundant power supply is now in operation.

Note: No change in switch device configuration is necessary for this installation.

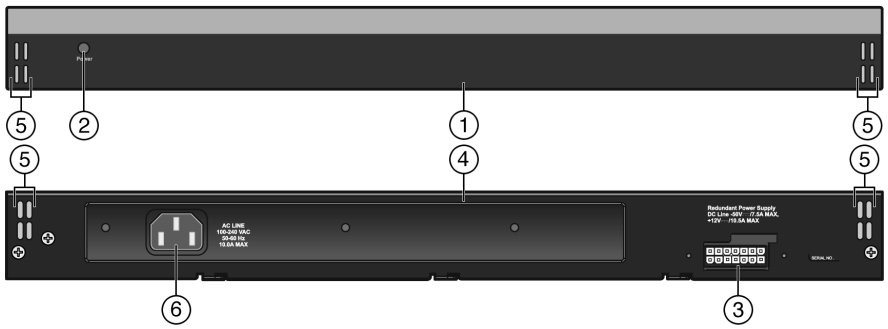
This completes the installation.

STK-RPS-500PS

The STK-RPS-500PS provides power backup to an Enterasys PoE-compliant B5 stackable switch. If for some reason the switch looses power from its internal power supply, the STK-RPS-500PS can provide up to 500 watts maximum operating power to support switch operation and the 48 Vdc necessary to support 48 Vdc/data connections to PDs (Powered Devices).

Figure 2-13 shows the front and rear view of the STK-RPS-500PS power supply.

Figure 2-13 STK-RPS-500PS (front and rear view)



1	STK-RPS-500PS (front view)	**4**	STK-RPS-500PS (rear view)
2	LED status indicator	**5**	Air vents for cooling
3	Redundant Power Supply connector	**6**	AC power input connector

The STK-RPS-500PS is shipped with the following:

- AC power cord
- RPS cable
- Four rubber feet (for flat surface installation)
- Two rack mount brackets
- Eight flathead screws (M3x6)

Installing the STK-RPS-500PS

You can install the STK-RPS-500PS on a flat surface or in a 19-inch rack.

Locate the STK-RPS-500PS within 152 cm (5 ft) of its power source.

Rack Mounting the STK-RPS-500PS

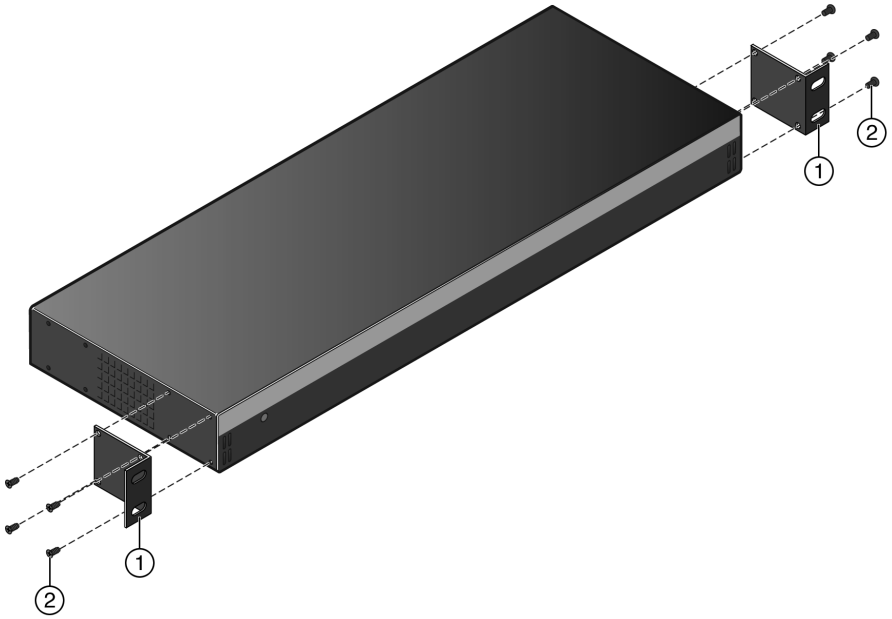
To install the STK-RPS-500PS in a 19-inch rack, you need:

- Two rackmount brackets and mounting screws (rackmount kit) shipped with the STK-RPS-500PS.
- Four customer-supplied screws to attach the STK-RPS-500PS to a standard 19-inch rack.

To install an STK-RPS-500PS in the shelf:

1. Attach the rackmount brackets to the STK-RPS-500PS, as shown in Figure 2-14, using the eight M3 x 6 mm flathead screws shipped with the STK-RPS-500PS.

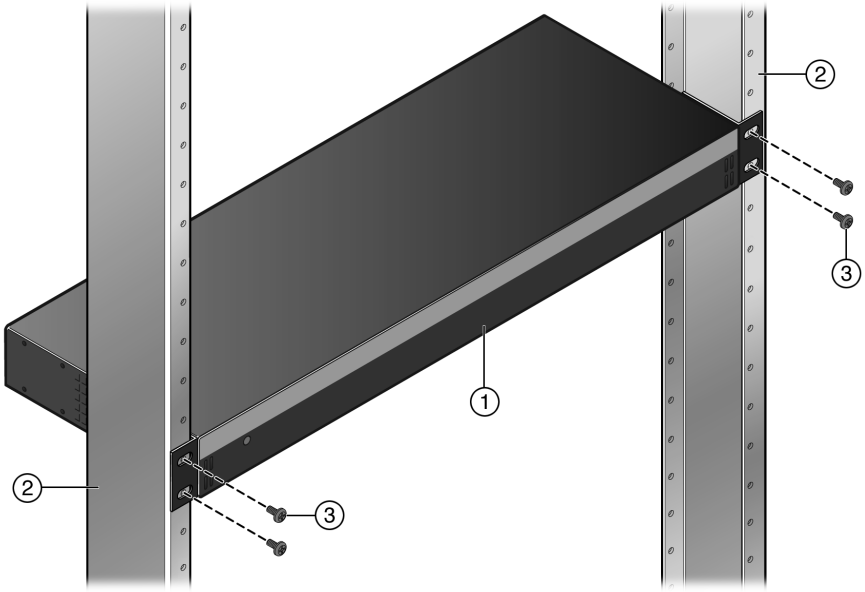
Figure 2-14 Attaching the Rackmount Brackets



1 Rackmount brackets (2) **2** M3 x 6 mm flathead screws (8)

2. With the mounting brackets attached, position the STK-RPS-500PS between the vertical frame members of the 19-inch rack as shown in Figure 2-15. Then fasten the STK-RPS-500PS securely to the frame using four customer-supplied rack screws.

Figure 2-15 Fastening the STK-RPS-500PS to the Rack



1 STK-RPS-500PS
2 Rails of 19-inch rack
3 Mounting screws

3. If you are installing the STK-RPS-500PS in a stacked configuration, repeat this procedure for each STK-RPS-500PS, then proceed to Connecting the RPS Cable and AC Power Cord.

Installing the STK-RPS-500PS on a Flat Surface

When installing the STK-RPS-500PS on a flat surface, the installation of the rubber feet is recommended to prevent the STK-RPS-500PS from sliding on a flat surface.

To install the rubber feet:

1. Place the STK-RPS-500PS on its back on a sturdy flat surface to gain access to the bottom of the chassis.
2. Remove the four rubber feet from their plastic bag in the shipping box.
3. Locate the four marked locations on the bottom four corners of the STK-RPS-500PS.
4. Remove the protective strip from the back of one rubber foot and position it on a marked location and press firmly into place. Repeat this procedure to install the remaining three rubber feet.
5. After installing the rubber feet, return the STK-RPS-500PS to its upright position.

Connecting the RPS Cable and AC Power Cord

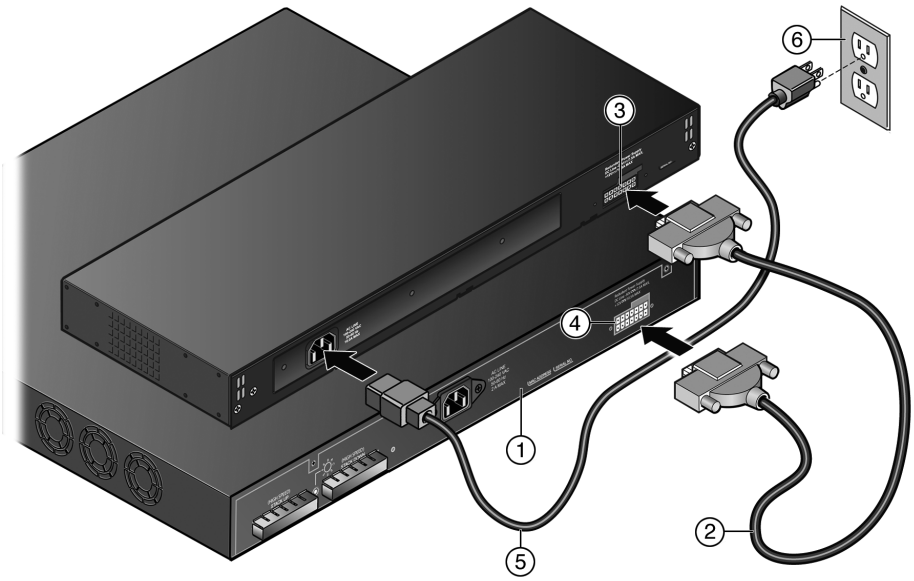
The STK-RPS-500PS is connected to the B5 PoE switch using the RPS cable, as follows:

1. Turn off the B5 switch and unplug it from the power source before installing the STK-RPS-500PS.

   The B5 switch does not support hot insertion/removal of an RPS cable. If you hot insert an RPS cable into a running B5 switch, the B5 switch will not recognize that an RPS cable is installed and the B5 switch will lose power if the primary power supply fails.

2. Connect one end of the RPS cable to the redundant power supply connector on the B5 switch. Then connect the other end to the redundant power supply connector at the rear of the STK-RPS-500PS as shown in Figure 2-16.

Figure 2-16 STK-RPS-500PS RPS Cable and AC Power Cord Connections



1	B5 switch	**5**	AC power cord (type varies depending on country)
2	RPS cable	**6**	AC power outlet with ground connection (type varies depending on country)
3	STK-RPS-500PS output connector		
4	B5 redundant power supply connector		

3. Connect a standard AC power cord to the AC input power connector on the STK-RPS-500PS, then plug the AC power cord into the main AC power outlet as shown in Figure 2-16.

   The AC power LED on the front of the STK-RPS-500PS will turn green to indicate a successful connection and the STK-RPS-500PS is operating properly.

4. Reconnect the B5 switch to the AC power source.

5. Turn on the B5 switch.

Connecting to the Console Port for Local Management

This section describes how to install an RS232 DTE interface cable to a PC, a VT series terminal, or a modem to a B5 for out-of-band sessions using CLI commands.

Note: When switches are connected in a stack configuration and all high-speed stacking cables are connected before powering up the switches, one switch in the stack will be automatically designated as the Manager of the stack and its Console port will remain active. All other switches will become Member switches and their Console ports will be deactivated.

What Is Needed

The following is a list of interface cables that may be needed to connect the DB9 male Console port connector on the switch. The cables are terminated by a DB9 female connector at one end, and by one of three type connectors at the other end, depending on the type connection needed for the remote device. The cables that may be needed are as follows:

- DB9 female-to-DB9 female (supplied with switch)
- DB9 female-to-DB25 female
- DB9 female-to-DB25 male

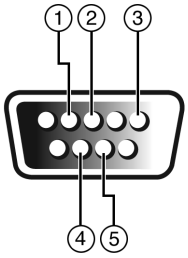
Using a DTE modem DB9 female-to-DB9 female cable, you can connect products equipped with a DB9 DTE male console port to an IBM or compatible PC running a VT series emulation software package.

Using a DTE modem DB9 female-to-DB25 female cable, you can connect products equipped with a DB9 DTE male console port to a VT series terminal or VT type terminals running emulation programs for the VT series.

Using a DTE modem DB9 female-to-DB25 male cable, you can connect products equipped with a DB9 DTE male console port to a Hayes compatible modem that supports 9600 baud.

The cable used must connect the Console port Received Data, Pin 2 to the Transmitted Data pin at the other end of the cable. The connection from the Console port Transmitted Data, Pin 3 (must be connected) to the Received Data pin cable connection at the other end of the cable. The DB9 Console port pin assignments are shown in Figure 2-17.

Figure 2-17 DB9 Male Console Port Pinout Assignments



1 Pin 2, Received Data (input)
2 Pin 3, Transmitted Data (output)
3 Pin 5, Signal Ground
4 Pin 7, Request to Send
5 Pin 8, Clear to Send
All other pins not connected.

Connecting to a PC

To connect a PC, running the VT terminal emulation, to a B5 Console port:

1. Connect the DB9 female connector at one end of the cable (supplied with switch) to the Console port on the B5 switch. (If there is a switch designated as the manager, connect to its Console port.)
2. Plug the DB9 female connector at the other end of cable into the communications port on the PC.
3. Turn on the PC and configure your VT emulation package with the following parameters:

Parameter	Setting
Mode	7 Bit Control
Transmit	Transmit=9600
Bits Parity	8 Bits, No Parity
Stop Bit	1 Stop Bit

When these parameters are set, the Startup screen will display. Proceed to "Connecting to the Network" on page 2-25.

Connecting to a VT Series Terminal

To connect a VT Series terminal to an B5 Console port, use a UTP serial interface cable terminated with a DB9 female connector and a DB25 female connector and proceed as follows:

1. Connect the DB9 female connector at one end of cable to the Console port DB9 male connector on the B5 switch. (If there is a switch designated as the manager, connect to its Console port.)
2. Plug the DB25 female connector at the other end of the cable into the port labeled COMM on the VT terminal.
3. Turn on the terminal and access the Setup Directory. Set the following parameters on your terminal:

Parameter	Setting
Mode	7 Bit Control
Transmit	Transmit=9600
Bits Parity	8 Bits, No Parity
Stop Bit	1 Stop Bit

When these parameters are set, the Startup screen will display. Proceed to "Connecting to the Network" on page 2-25.

Connecting to a Modem

To connect a modem to a B5 Console port, use a UTP serial interface cable terminated with a DB9 female connector and a DB25 male connector and proceed as follows:

1. Connect the DB9 female connector at one end of the cable to the Console port DB9 connector on the B5 switch. (If there is a switch designated as the manager, connect to its Console port.)
2. Plug the DB25 male connector at the other end of the cable into the modem communications port.
3. Turn on the modem and make sure the remote modem is ON.
4. With your PC connected to the remote modem, configure your VT emulation package with the following parameters:

Parameter	Setting
Mode	7 Bit Control
Transmit	Transmit=9600
Bits Parity	8 Bits, No Parity
Stop Bit	1 Stop Bit

When these parameters are set, the Startup screen will display. If the switches are in a stacked configuration, proceed to "Connecting to the Network" on page 2-25.

Connecting to the Network

The following procedures cover the cable connections from the network or other devices to the B5 switch front panel ports.

- "Connecting UTP Cables to RJ45 Ports" on page 2-26
- "Installing an Optional SFP/10G Transceiver" on page 2-27
- "Connecting Fiber-Optic Cables to SFP/10G Ports" on page 2-29

Connecting UTP Cables to RJ45 Ports

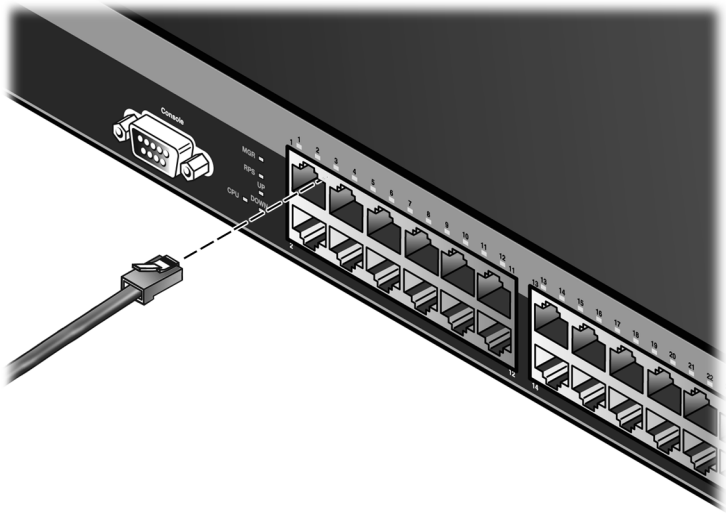
The fixed RJ45 front panel 10/100/1000 Mbps ports support Auto MDIX, which means that you can use straight-through or crossover twisted pair cabling.

Note: All fixed RJ45 front panel ports support Category 5 Unshielded Twisted Pair (UTP) cabling with an impedance between 85 and 111 ohms. Category 3 cable may be used if the connection is going to be used only for 10 Mbps.

To connect twisted pair segments to the switch, refer to Figure 2-18 and proceed as follows:

1. Ensure that the device to be connected at the other end of the segment is powered on.
2. Connect the twisted pair segment to the switch by inserting the RJ45 connector on the twisted pair segment into the desired RJ45 port on the B5 switch.

Figure 2-18 Connecting a UTP Cable Segment to an RJ45 Port



3. Verify that a link exists by checking that the Link/Activity LED is on (solid green or blinking green). If the Link/Activity LED is off, perform the following steps until it is on:
   a. Verify that the cabling being used is Category 5 or better with an impedance between 85 and 111 ohms with a maximum length of 100 meters (328 feet).
   b. Verify that the device at the other end of the twisted pair segment is on and properly connected to the segment.
   c. Verify that the RJ45 connectors on the twisted pair segment have the proper pinouts and check the cable for continuity.
4. If a link is not established, contact Enterasys Networks. Refer to “Getting Help” on page xvii for details.

Repeat all steps above until all connections have been made.

Preparing to Install an Optional SFP/10G Transceiver

Before removing the pluggable transceiver from the antistatic packaging, Enterasys Networks recommends that you use an antistatic wrist strap (not supplied) to prevent damage from static discharge. To use the wrist strap, refer to the instructions provided with your antistatic wrist strap.

1. Remove the pluggable transceiver from its packaging.
2. If there is a protective dust cover in the pluggable transceiver connector, do not remove it at this time.

Installing an Optional SFP/10G Transceiver



Note: Each combo SFP port in use eliminates the availability of one RJ45 port. In other words, only 24 or 48 ports can be active at any given time on B5 switches equipped with a combination of RJ45 and SFP interfaces.

When an SFP transceiver in SFP port 23 establishes a link, RJ45 port 23 is disabled. When an SFP transceiver in SFP port 24 establishes a link, RJ45 port 24 is disabled.

This section describes how to install an SFP or 10G optical transceiver into the following B5 ports:

Transceiver Type	Switch	Ports
SFP	B5G124-24 and B5G124-24P2	Combo ports 23 through 26
	B5G124-48 and B5G124-48P2	Combo ports 47 through 50
	B5K125-24 and B5K125-24P2	Combo ports 23 and 24
	B5K125-48 and B5K125-48P2	Combo ports 47 and 48
10G	B5K125-24 and B5K125-24P2	10G ports 25 and 26
	B5K125-48 and B5K125-48P2	10G ports 49 and 50



Warning: Fiber-optic SFP/10Gs use Class 1 lasers. Do not use optical instruments to view the laser output. The use of optical instruments to view laser output increases eye hazard. When viewing the output optical port, power must be removed from the network adapter.

Advertencia: Los SFP/10Gs de fibra optica usan lasers de clase 1. No se debe usar instrumentos opticos para ver la potencia laser. El uso de los instrumentos opticos para ver la potencia laser incrementa el riesgo a los ojos. Cuando vean el puerto de la potencia optica, la corriente debe ser removida del adaptador de la red.

Warnhinweis: SFP/10Gs mit Fiber-Optik Technologie benutzen Laser der Klasse 1. Benutzen sie keinesfalls optische Hilfsmittel, um die Funktion des Lasers zu überprüfen. Solche Hilfsmittel erhöhen die Gefahr von Sehschäden. Wenn sie den optischen Port überprüfen möchten stellen Sie sicher, dass die Komponente von der Spannungsversorgung getrennt ist.

Caution: Carefully follow the instructions in this manual to avoid damaging the SFP/10G and switch equipment.

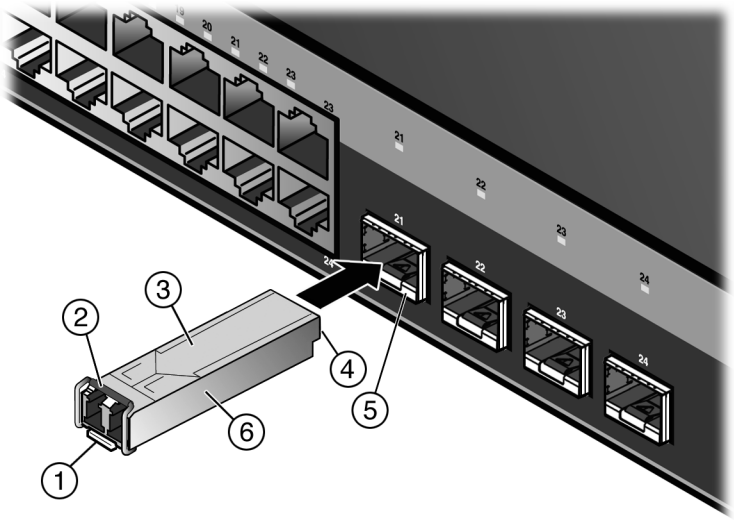
The SFP/10G and switch equipment are sensitive to static discharges. Use an antistatic wrist strap and observe all static precautions during this procedure. Failure to do so could result in damage to the SFP/10G and switch equipment. Always leave the SFP/10G in the antistatic bag or an equivalent antistatic container when not installed.

Precaución: Siga las instrucciones del manual para no dañar el SFP/10G ni del aparato, puesto que son muy sensible a las descargas de electricidad estática. Utilice la pulsera antiestática y tome todas las precauciones necesarias durante este procedimiento. Si no lo hace, podría dañar el SFP/10G o del aparato. Mientras no esté instalado, mantenga el SFP/10G en su bolsa antiestática o en cualquier otro recipiente antiestático.

To install an SFP/10G transceiver, refer to Figure 2-19 and proceed as follows.

1. With an antistatic wrist strap attached to your wrist, remove the transceiver from its packaging. If there is a protective dust cover in the transceiver connector, do not remove it at this time.
2. Hold the transceiver so that the connector will seat properly.
3. Carefully align the transceiver with the port slot as shown in Figure 2-19.
4. Push the transceiver into the port slot until it “clicks” and locks into place.

Figure 2-19 Installing an SFP/10G Transceiver (shown with LC connector and without dust cover)



1	Release tab	**4**	Edge connector (insertion side)
2	Wire handle	**5**	Port slot
3	Transceiver top side	**6**	Transceiver side view

Removing an SFP/10G Transceiver

Caution: Do NOT remove the SFP/10G from the port slot without releasing it. The SFP/10G is released by pulling down on its wire handle. Attempting to remove the SFP/10G without releasing it can damage the SFP/10G.

The SFP/10G and its host switch are sensitive to static discharges. Use an antistatic wrist strap and observe all static precautions during this procedure. Failure to do so could result in damaging the SFP/10G or host switch. Always leave the SFP/10G in the antistatic bag or an equivalent antistatic container when not installed.

Precaución: NO quite el SFP/10G de la ranura sin antes abrir la traba ubicada en la parte frontal del SFP/10G. Si lo hace, puede dañar el SFP/10G, puesto que es muy sensible a las descargas de electricidad estática, al igual que el dispositivo host. Utilice la pulsera antiestática y tome todas las precauciones necesarias durante este procedimiento. Si no lo hace, pude dañar el SFP/10G o el dispositivo host. Mientras no esté instalado, mantenga el SFP/10G en su bolsa antiestática o en cualquier otro recipiente antiestático.

To remove a transceiver from a port slot, refer back to Figure 2-19 on page 2-28 and proceed as follows:

1. With an antistatic wrist strap attached to your wrist, remove the cables connected to the transceiver.
2. Release the transceiver using its wire handle. Specific operation and location of the handle will vary depending on transceiver type.
3. Grasp the sides of the transceiver and pull it straight out of the port slot.

If storing or shipping an SFP/10G, which has a fiber-optic connector, insert its protective dust cover to protect the ends of the fiber-optic fibers from dust or contamination.

Connecting Fiber-Optic Cables to SFP/10G Ports

Before connecting cables to SFP/10G ports, you must install the appropriate transceiver as described in "Installing an Optional SFP/10G Transceiver" on page 2-27. This section describes how to connect a 1- or 10-Gigabit fiber-optic segment from the network or other devices to an SFP or 10G port connector (LC or MT-RJ).

Each fiber-optic link consists of two fiber-optic strands within the cable for Transmit (TX) and Receive (RX). The transmit strand from a device port connects to the receive port of a fiber-optic 1-/10- Gigabit Ethernet device at the other end of the segment. The receive strand of the applicable LC or MT-RJ port connects to the transmit port of the fiber-optic 1-/10- Gigabit Ethernet device.

Caution: Do not touch the ends of the fiber-optic strands, and do not let the ends come in contact with dust, dirt, or other contaminants. Contamination of cable ends causes problems in data transmissions. If the ends of the fiber-optic strands become contaminated, use a canned duster to blow the surfaces clean. A fiber-port cleaning swab saturated with optical-grade isopropyl alcohol may also be used to clean the ends.

Precaución: No toque los extremos de los cables de fibra óptica y evite su contacto con el polvo, la suciedad o con cualquier otro contaminante. Si los extremos de los cables se ensucian, es posible que la transmisión de datos se vea afectada. Si nota que losextremos de los cables de fibra óptica se ensucian, utilice aire comprimido paralimpiarlos. También puede limpiarlos con un estropajo embebido en alcohol isopropílico.

Refer to Figure 2-20 as you perform the following procedure.

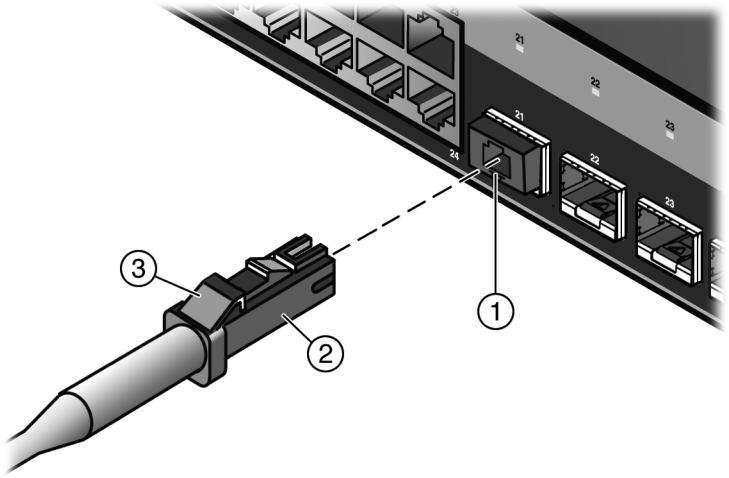
To connect an LC or MT-RJ cable connector to an SFP or 10G port connector:

1. Remove the protective covers (not shown) from the uplink port SFP/10G and from the connectors on each end of the cable.

Note: Leave the protective covers in place when the connectors are not in use to prevent contamination.

2. Insert the cable connector into the SFP/10G connector until it clicks into place.

Figure 2-20 Cable Connection (LC shown) to Uplink Port with SFP/10G Installed



1 Combo SFP port with transceiver installed **2** LC cable connector **3** Release tab

3. Plug the other end of the cable into the appropriate port on the other device. Some cables may be terminated at the other end with two separate connectors, one for each fiber-optic strand. In this case, ensure that the transmit fiber-optic strand from the B5 switch is connected to the receive port of the other device, and the receive fiber-optic strand on the B5 switch is connected to the transmit port of the other device.
4. Repeat this procedure for other SFP/10G ports, if needed.
5. If an SFP/10G port is unused, install a dust cover.

Completing the Installation

After installing the switch and making the connections to the network, access the switch management, as described below.

Initial Logon to Switch Management

To initially access switch management from your local PC, terminal, or modem connection, proceed as follows at the displayed startup screen:

1. Enter **rw** (Read-Write) for Username.
2. At the Password prompt, press **Enter** (RETURN).

3. For details on how to configure the switch using the command line interface, refer to the *Enterasys B5 CLI Reference*. The CLI commands enable you to set a new password and perform more involved management configurations on the switch. One of many capabilities is to reconfigure the stack, reassigning the Manager function to another switch and configuring the order of Member switches that will take over the stack management functions in case the operating Manager is powered down, malfunctions, or is removed from the stack.
4. After the initial configuration, you can also use WebView (Enterasys Networks' embedded web server) for configuration and management tasks.

   All the guides associated with the switch are available online at:
   http://www.enterasys.com/support/manuals

Once you are confident that the installation is successful, route and secure your cables. If you require assistance, contact Enterasys Networks using one of the methods described in "Getting Help" on page xvii.

3

Troubleshooting

This chapter contains instructions on troubleshooting the B5 switch as required. This can include:

For information about...	Refer to page...
Checking the LEDs	3-2
Troubleshooting Checklist	3-8
Using the Password Reset Button	3-9
Removing the Switch from a Rack	3-9



Electrical Hazard: Only qualified personnel should install or service this unit.

Riesgo Eléctrico: Nada mas personal capacitado debe de instalar o darle servicio a esta unida.

Elektrischer Gefahrenhinweis: Installationen oder Servicearbeiten sollten nur durch ausgebildetes und qualifiziertes Personal vorgenommen werden.



Warning: Do not connect or disconnect any connections while circuit is live, unless area is known to be non-hazardous. Secure any external connections that mate to this equipment by using the screws, safety bars, or other means provided with this equipment.

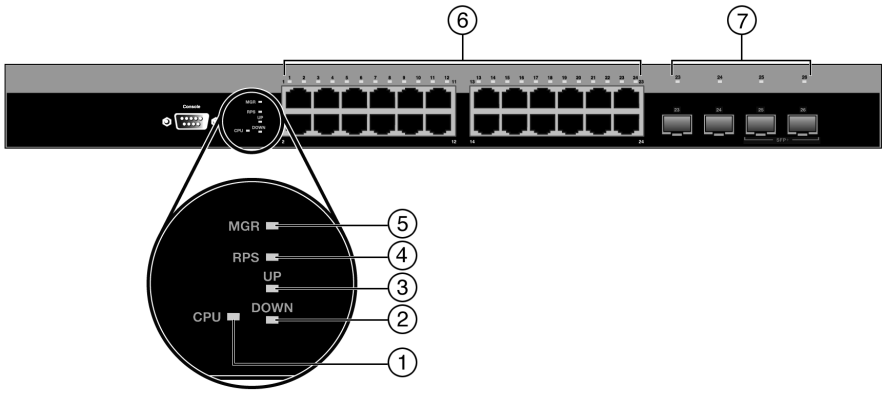
Advertencia: No conecte ni desconecte ninguna conexión mientras el circuito tenga corriente, a menos que esté seguro de que el área no es peligrosa. Asegure cualquier conexión externa que se una a este equipo usando tornillos, barras de seguridad u otros medios que se proporcionen con el mismo.

Warnhinweis: Ist der Stromkreis in Betrieb dürfen keine Verbindungen getrennt oder hergestellt werden, es sei denn, die Umgebung gilt als ungefährlich. Alle externen Verbindungen zu diesem Gerät müssen mithilfe von Schrauben, Sicherheitsvorrichtungen o. ä. gesichert werden.

Checking the LEDs

The following sections define the behavior of the LEDs on the B5 chassis models. Refer to Figure 3-1, Figure 3-2, Figure 3-3, and Figure 3-4 for the location of the LEDs on the chassis.

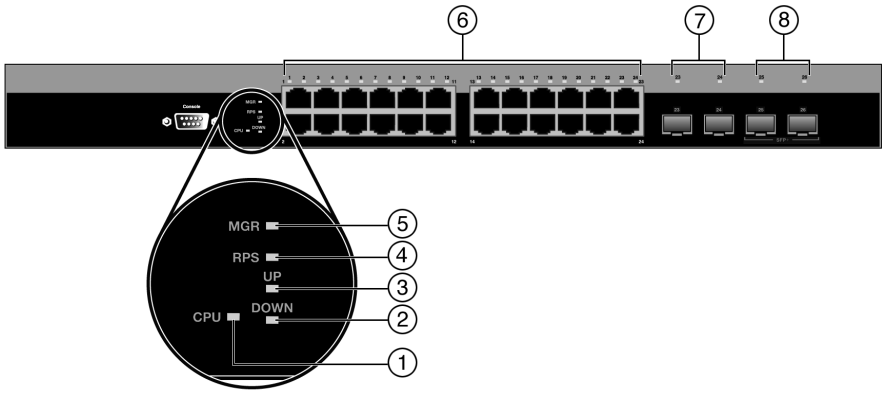
Figure 3-1 B5G124-24 and B5G124-24P2 LEDs



1 CPU LED
2 STACK DOWN LED
3 STACK UP LED
4 RPS LED for redundant power-source status
5 MGR LED
6 Link/Activity/PoE LEDs for 10/100/1000 Mbps for RJ45 ports 1 through 24
7 Link/Activity LEDs for SFP ports 23 through 26[1]

1. SFP ports 23 and 24 will only establish a link when the RJ45 port equivalent is not linked. (For example, when SFP port 23 is linked, RJ45 port 23 is deactivated. When the SFP is not linked, the RJ45 port 23 is reactivated and can establish a link as long as the SFP port 23 is not linked first.)

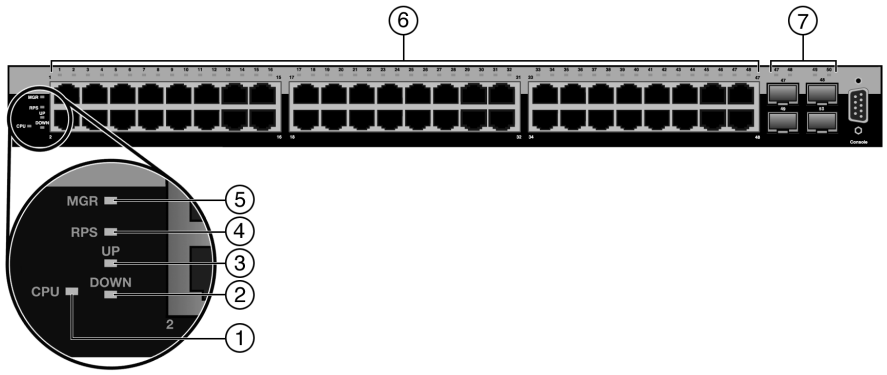
Figure 3-2 B5K125-24 and B5K125-24P2 LEDs



1 CPU LED
2 STACK DOWN LED
3 STACK UP LED
4 RPS LED for redundant power-source status
5 MGR LED
6 Link/Activity/PoE LEDs for 10/100/1000 Mbps for RJ45 ports 1 through 24
7 Link/Activity LEDs for SFP ports 23 and 24[1]
8 Link/Activity LEDs for 10G ports 25 and 26

1. These SFP ports will only establish a link when the RJ45 port equivalent is not linked. (For example, when SFP port 23 is linked, RJ45 port 23 is deactivated. When the SFP is not linked, the RJ45 port 23 is reactivated and can establish a link as long as the SFP port 23 is not linked first.)

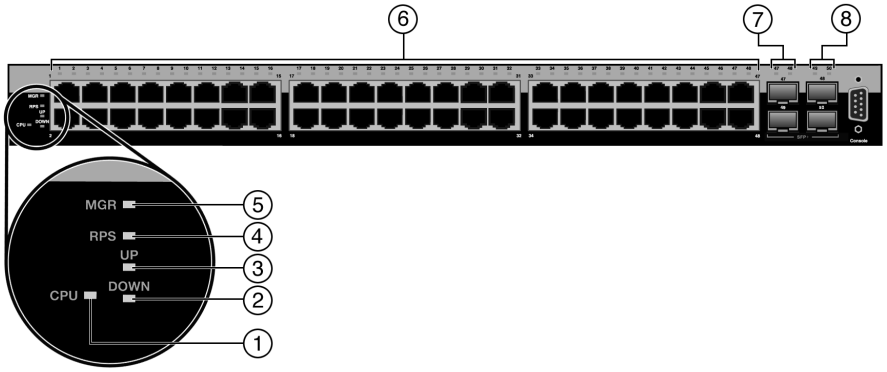
Figure 3-3 B5G124-48 and B5G124-48P2 LEDs



1 CPU LED
2 STACK DOWN LED
3 STACK UP LED
4 RPS LED for redundant power source status
5 MGR LED
6 Link/Activity/PoE LEDs for 10/100/1000 Mbps RJ45 ports 1 through 48
7 Link/Activity LEDs for SFP ports 47 through 50 [1]

1. These SFP ports will only establish a link when the RJ45 port equivalent is not linked. (For example, when SFP port 47 is linked, RJ45 port 47 is deactivated. When the SFP is not linked, the RJ45 port 47 is reactivated and can establish a link as long as the SFP port 47 is not linked first.)

Figure 3-4 B5K125-48 and B5K125-48P2 LEDs



1 CPU LED
2 STACK DOWN LED
3 STACK UP LED
4 RPS LED for redundant power source status
5 MGR LED
6 Link/Activity/PoE LEDs for 10/100/1000 Mbps RJ45 ports 1 through 48
7 Link/Activity LEDs for SFP ports 47 and 48[1]
8 Link/Activity LEDs for 10G ports 49 and 50

1. These SFP ports will only establish a link when the RJ45 port equivalent is not linked. (For example, when SFP port 47 is linked, RJ45 port 47 is deactivated. When the SFP is not linked, the RJ45 port 47 is reactivated and can establish a link as long as the SFP port 47 is not linked first.)

MGR LED

The MGR LED indicates whether the switch is operating as a member or as the manager of the stack, as described in Table 3-1.

Table 3-1 MGR LED Definitions

Color	State	Recommended Action
Off	Switch is operating as a member in the stack.	None.
Green	**Solid.** Switch is operating as the manager of the stack.	None.

RPS LED

The RPS LED indicates the state of the internal power supply and whether a redundant power supply is providing power, as described in Table 3-2.

Table 3-2 RPS LED Definitions

Color	State	Recommended Action
Off	Connected RPS not in use.	None.
	Internal power supply failed.	If an RPS is connected to the switch, it should be providing power. Perform the following: 1. Ensure the RPS is powered on. 2. Ensure that the AC power cord to the RPS is plugged in correctly and that there is power at the AC power source. 3. Replace the power cord with a known good one. 4. Ensure the DC power cord from the RPS to the switch is plugged in correctly. 5. If the problem persists, contact Enterasys Networks for technical support.
Amber	**Solid.** The switch internal power supply failed, and the RPS is providing the proper power to the switch.	None.

UP LED

The STACK UP LED indicates the state of the connection to the switch stack down connector, as described in Table 3-3.

Table 3-3 UP LED Definitions

Color	State	Recommended Action
Off	No valid connection to switch stack down connector.	1. Make sure the switch connected to the stack down connector is powered on. 2. Replace cable with a known good one. 3. If the problem still exists, contact Enterasys Networks for technical support.
Green	**Solid.** Valid connection to switch stack down connector.	None.
	Blinking. Information is being transferred through the high-speed stacking cable.	None.

DOWN LED

The STACK DOWN LED indicates the state of the connection to the switch stack up connector, as described in Table 3-4.

Table 3-4 DOWN LED Definitions

Color	State	Recommended Action
Off	No valid connection to switch stack up connector.	1. Make sure the switch connected to the stack up connector is powered on. 2. Replace cable with a known good one. 3. If the problem still exists, contact Enterasys Networks for technical support.
Green	**Solid.** Valid connection to switch stack up connector.	None.
	Blinking. Information is being transferred through the high-speed stacking cable.	None.

CPU LED

The CPU LED indicates the status of the system operation, as described in Table 3-5.

Table 3-5 CPU LED Definitions

Color	State	Recommended Action
Off	Power disconnected.	If the LED remains off after connecting AC power or DC power from an RPS, check the following: 1. Make sure there is power at the power source (AC source and RPS if applicable). 2. Replace the power cord with a known good one or check the continuity of the power cord. 3. If the switch still does not power up, the system may have a fatal error. Contact Enterasys Networks for technical support.
Red	**Solid.** Boot-up failed.	If the LED remains red for several minutes, the system may have a fatal error. Contact Enterasys Networks for technical support.
Amber	**Solid.** Diagnostics are running.	If the LED remains amber for more than several minutes, contact Enterasys Networks for technical support.
	Blinking. Code image is being downloaded.	None.
Green	**Solid.** System is fully functional.	None.
	Blinking. Boot-up in process.	None.

Link/Activity LEDs

Table 3-6 describes the status of RJ45 port LEDs. Table 3-7 describes the status of RJ45 PoE LEDs. Table 3-8 describes the status of SFP/10G port LEDs.

Table 3-6 RJ45 Port LED Definitions

Color	State	Recommended Action
Off	No link.	None.
	No activity or port in standby.	None.
	If the port is one of the highest numbered last four RJ45 ports, the port is deactivated because the SFP/10G port with the same port number is linked.	None.
Green	**Solid**. Port is linked, but the interface is not receiving any traffic.	None.
	Blinking. Port is linked and traffic is being received or transmitted by the interface.	None.

Table 3-7 RJ45 PoE Port LED Definitions

Color	State	Recommended Action
Off	No link.	None.
	No activity or port in standby.	None.
	If the port is one of the highest numbered last four RJ45 ports, the port is deactivated because the SFP/10G port with the same port number is linked.	None.
Green	**Solid.** Port is linked, but the interface is not receiving any traffic. PoE power is available.	None.
	Blinking. Port is linked and traffic is being received or transmitted by the interface. PoE power is available.	None.
Amber	**Solid.** Port is linked, but traffic is not being received or transmitted by the interface. PoE power has failed. PoE power may have been deactivated because of excessive power requirement or deactivated by configuration. Power supply failed.	1. Check the attached powered device to see if it is defective. 2. Internal power supply failed, contact Enterasys Networks for technical support.
	Blinking. Port is linked and traffic is being received or transmitted by the interface. PoE power has failed. PoE power may have been deactivated because of excessive power requirement or deactivated by configuration. Power supply failed.	1. Check the attached powered device to see if it is defective. 2. Internal power supply failed, contact Enterasys Networks for technical support.

Table 3-8 SFP/10G Port LED Definitions

Color	State	Recommended Action
Off	No link.	None.
	No activity or port in standby.	None.
	The port is deactivated because the RJ45 port with the same port number is linked.	None.
Green	**Solid**. Port is linked, but the interface is not receiving any traffic.	None.
	Blinking. Port is linked and traffic is being received or transmitted by the interface.	None.

Troubleshooting Checklist

If the device is not working properly, refer to Table 3-9 for a checklist of problems, possible causes, and recommended actions to resolve the problem.

Table 3-9 Troubleshooting Checklist

Problem	Possible Cause	Recommended Action
All LEDs are OFF.	Loss of power.	Ensure that the switch was installed properly according to the installation instructions in Chapter 2, **Installation**. Ensure that the power cords are connected properly to an active power source that meets the AC input specifications for this switch. Ensure that the AC power cord is not damaged. Replace the power cord with a known good one.
No local management startup screen.	Incorrect terminal setup.	Refer to the *Enterasys B5 CLI Reference* for proper setup procedures.
	Improper Console cable pinouts.	Refer to Appendix A, **Specifications**, for proper Console port pinouts.
	Corrupt firmware image, or hardware fault.	If possible, attempt to download the image to the device again. Refer to the *Enterasys B5 CLI Reference* for details.
Cannot navigate beyond startup screen.	Improper Community Names Table.	Refer to the *Enterasys B5 CLI Reference* for the Community Names Table setup.
Cannot contact the switch through in-band management.	IP address not assigned.	Refer to the *Enterasys B5 CLI Reference* for the IP address assignment procedure.
	Port is disabled.	This is normal if the switch is a Member in a stack. If operating as a standalone switch or as the Manager in a stack, enable the port. Refer to the *Enterasys B5 CLI Reference* for instructions to enable/disable ports.
	No link to device.	Verify that all network connections between the network management station and the device are valid and operating. If the problem continues, contact Enterasys Networks for technical support.
Port(s) goes into standby for no apparent reason.	Loop condition detected.	Verify that Spanning Tree is enabled. Refer to the *Enterasys B5 CLI Reference* for the instructions to set the type of STA. Review the network design and delete unnecessary loops. If the problem continues, contact Enterasys Networks for technical support.
User parameters (IP address, device and module name, etc.) were lost when the switch power was cycled.	Clear NVRAM was set using CLI commands.	Reenter the lost parameters as necessary. Refer to the *Enterasys B5 CLI Reference* for the instructions to configure the switch using CLI commands. If the problem continues, contact Enterasys Networks for technical support.

Using the Password Reset Button

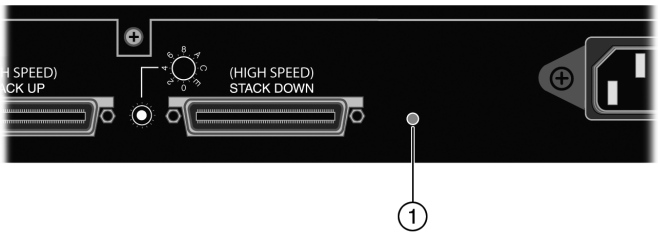
If you forget the B5 login password, use the password reset button to reset the password to the default value as described in the following procedure.

Note: Notify the system manager before changing the password.

To reset the B5 password:

1. Locate the password reset button on the back of the switch as shown in Figure 3-5.

Figure 3-5 Locating the Password Reset Button



1 Password reset button

2. Press-and-hold the password reset button while the switch is operational. This changes the login password to the default password and will be indicated by means of the command line interface (CLI) only.
3. You can now logon to the switch using the default password via the Console port and assign a new password using the CLI.
4. To access switch management from your local PC, terminal, or modem connection, refer to the *Enterasys B5 CLI Reference* for instructions on how to log in and enter a new password.

If you require assistance, contact Enterasys Networks using one of the methods described in "Getting Help" on page xvii.

Note: If the admin user account is locked out (typically because someone entered the wrong password multiple times), resetting the admin password with the password reset button will not unlock the admin user account. In order to unlock the admin user account, you can wait for the configured lockout time to expire or you can power cycle the switch to reboot it.

Removing the Switch from a Rack

To remove the B5 switch from a rack:

1. While supporting the switch so it does not fall, carefully remove the mounting screws from the two brackets that attach the switch to the rack.
2. If necessary, remove each bracket from the switch by removing the mounting kit screws as shown in Figure 2-3 on page 2-5.

A

Specifications

This appendix provides the following information:

For information about...	Refer to page...
Switch Specifications	A-1
RPS Specifications	A-5
Torque Values	A-9
Pluggable Transceiver Specifications	A-9
Console Port Pinout Assignments	A-9
Regulatory Compliance	A-9

Enterasys Networks reserves the right to change specifications at any time without notice.

Switch Specifications

Table A-1 provides the I/O ports, processors and memory, physical, and environmental specifications for the B5 models.

Table A-1 B5 Switch Specifications

Item	Specification
B5G124-24 and B5G124-24P2	
RJ45 ports 1 through 24	Twenty-four 10BASE-T/100BASE-TX/1000BASE-T compliant ports with auto-sensing and auto-negotiation via RJ45 connectors. These ports also support 802.3at and 802.3af PoE connections on the B5G124-24P2.
Combo SFP ports 23 through 26	Four ports that support optional pluggable transceiver 100BASE-FX, 1000BASE-SX, 1000BASE-LX, and 1000BASE-TX fiber-optic connections. When an SFP transceiver in SFP port 23 establishes a link, RJ45 port 23 is disabled. When an SFP transceiver in SFP port 24 establishes a link, RJ45 port 24 is disabled.
B5G124-48 and B5G124-48P2	
RJ45 ports 1 through 48	Forty-eight 10BASE-T/100BASE-TX/1000BASE-T compliant ports with auto-sensing and auto-negotiation via RJ45 connectors. These ports also support 802.3at and 802.3af PoE connections on the B5G124-48P2.

Table A-1 B5 Switch Specifications (continued)

Item	Specification
Combo SFP ports 47 through 50	Four ports that support optional pluggable transceiver 100BASE-FX, 1000BASE-SX, 1000BASE-LX, and 1000BASE-TX fiber-optic connections. When an SFP transceiver in SFP port 47 establishes a link, RJ45 port 47 is disabled. When an SFP transceiver in SFP port 48 establishes a link, RJ45 port 48 is disabled.
B5K125-24 and B5K125-24P2	
RJ45 ports 1 through 24	Twenty-four 10BASE-T/100BASE-TX/1000BASE-T compliant ports with auto-sensing and auto-negotiation via RJ45 connectors. These ports also support 802.3at and 802.3af PoE connections on the B5K125-24P2.
Combo SFP ports 23 and 24	Two ports that support optional pluggable transceiver 100BASE-FX, 1000BASE-SX, 1000BASE-LX, and 1000BASE-TX fiber-optic connections. When an SFP transceiver in SFP port 23 establishes a link, RJ45 port 23 is disabled. When an SFP transceiver in SFP port 24 establishes a link, RJ45 port 24 is disabled.
10G ports 25 and 26	Two ports that support 1000BASE-X copper and 10GBASE-SR/LR/LRM pluggable transceiver connections.
B5K125-48 and B5K125-48P2	
RJ45 ports 1 through 48	Forty-eight 10BASE-T/100BASE-TX/1000BASE-T compliant ports with auto-sensing and auto-negotiation via RJ45 connectors. These ports also support 802.3at and 802.3af PoE connections on the B5K125-48P2.
Combo SFP ports 47 and 48	Two ports that support optional pluggable transceiver 100BASE-FX, 1000BASE-SX, 1000BASE-LX, and 1000BASE-TX fiber-optic connections. When an SFP transceiver in SFP port 47 establishes a link, RJ45 port 47 is disabled. When an SFP transceiver in SFP port 48 establishes a link, RJ45 port 48 is disabled.
10G ports 49 and 50	Two ports that support 1000BASE-X copper and 10GBASE-SR/LR/LRM pluggable transceiver connections.
Processors/Memory	
Processor	MPC8378, 667 MHz processor
Synchronous Dynamic Random Access Memory (SDRAM)	512 MB
FLASH Memory	32 MB

Table A-1 B5 Switch Specifications (continued)

Item	Specification
Performance	
Throughput capacity wirespeed Mpps (switch/stack)	• B5G124-24 — 35.7 Mpps / 85.7 Mpps • B5G124-24P2 — 35.7 Mpps / 85.7 Mpps • B5G124-48 — 71.4 Mpps /571.4 Mpps • B5G124-48P2 — 71.4 Mpps /571.4 Mpps • B5K125-24 — 65.5 Mpps /523.8 Mpps • B5K125-24P2 — 65.5 Mpps /523.8 Mpps • B5K125-48 — 101.2 Mpps /809.5 Mpps • B5K125-48P2 — 101.2 Mpps /809.5 Mpps
Switching capacity (switch/stack)	• B5G124-24 — 48 Gbps /384 Gbps • B5G124-24P2 — 48 Gbps /384 Gbps • B5G124-48 — 96 Gbps /768 Gbps • B5G124-48P2 — 96 Gbps /768 Gbps • B5K125-24 — 88 Gbps /704 Gbps • B5K125-24P2 — 88 Gbps /704 Gbps • B5K125-48 — 136 Gbps /1,088 Gbps • B5K125-48P2 — 136 Gbps /1,088 Gbps
Stacking capacity(switch/stack)	• B5G124-24 — 52 Gbps /416 Gbps • B5G124-24P2 — 52 Gbps /416 Gbps • B5G124-48 — 52 Gbps /416 Gbps • B5G124-48P2 — 52 Gbps /416 Gbps • B5K125-24 — 52 Gbps /416 Gbps • B5K125-24P2 — 52 Gbps /416 Gbps • B5K125-48 — 52 Gbps /416 Gbps • B5K125-48P2 — 52 Gbps /416 Gbps
Aggregate throughput capacity (switch/stack)	• B5G124-24 — 100 Gbps /800 Gbps • B5G124-24P2 — 100 Gbps /800 Gbps • B5G124-48 — 148 Gbps /1,184 Gbps • B5G124-48P2 — 148 Gbps /1,184 Gbps • B5K125-24 — 140 Gbps /1,120 Gbps • B5K125-24P2 — 140 Gbps /1,120 Gbps • B5K125-48 — 188 Gbps /1,504 Gbps • B5K125-48P2 — 188 Gbps /1,504 Gbps
Physical	
Dimensions	44.1 W x 4.4 H x 36.85 D cm 17.4 W x 1.7 H x 14.5 D in
Approximate Weight	Gross: 11.79 kg (26 lb)
Mean Time Between Failure (MTBF)	Refer to the MTBF web site at URL http://www.enterasys.com/support/mtbf/

Table A-1 B5 Switch Specifications (continued)

Item	Specification
Power Specifications	
Input Voltage	100 to 240 VAC
Power Consumption	• B5G124-24 — 63W (214 BTU/hr) • B5G124-24P2 — 130W (444 BTU/hr) • B5G124-48 — 76W (258 BTU/hr) • B5G124-48P2 — 225W (768 BTU/hr) • B5K125-24 — 59W (200 BTU/hr) • B5K125-24P2 — 136W (463 BTU/hr) • B5K125-48 — 94W (321 BTU/hr) • B5K125-48P2 — 228W (777 BTU/hr)
Input Frequency	50 to 60 Hz
Input Current	• B5G124-24 — 2 A Max • B5G124-24P2 — 7.5 A Max • B5G124-48 — 2 A Max • B5G124-48P2 — 7.5 A Max • B5K125-24 — 2 A Max • B5K125-24P2 — 7.5 A Max • B5K125-48 — 2 A Max • B5K125-48P2 — 7.5 A Max
Thermal Output	Same as Power Consumption (above)
Environmental	
Operating Temperature	0°C to 50°C (32°F to 122°F)
Storage Temperature	-40°C to +70°C (-40°F to 158°F)
Operating Relative Humidity	5% to 95% (non-condensing)
Altitude	Operational up to 10,000 feet Non-operationally tested to 15,000 feet

RPS Specifications

STK-RPS-150CH2 Chassis Specifications

Table A-2 provides the physical specifications for the STK-RPS-150CH2.

Table A-2 STK-RPS-150CH2 Specifications

Item	Specification
Power supply slots	Two slots for optional STK-RPS-150PS power supplies
Dimensions without mounting brackets	5.5 H x 48.2 W x 18.0D (cm) 2.2 H x 19.0 W x 7.0 D (in.)
Net Weight (Unit Only)	0.95 kg (2.09 lb)
Gross Weight (Packaged Unit)	1.60 kg (3.52 lb)

STK-RPS-150CH8 Chassis Specifications

Table A-3 provides the physical specifications for the STK-RPS-150CH8.

Table A-3 STK-RPS-150CH8 Specifications

Item	Specification
Power supply slots	Eight slots for optional STK-RPS-150PS power supplies
Dimensions without mounting brackets	22.26 H x 44.0 W x 26.4 D (cm) 8.77 H x 17.3 W x 10.4 D (in.)
Net Weight (Unit Only)	5.27 kg (11.6 lb)
Gross Weight (Packaged Unit)	6.5 kg (14.3 lb)

STK-RPS-150PS Specifications

Table A-4 provides the specifications for the STK-RPS-150PS.

Table A-4 STK-RPS-150PS Specifications

Item	Specification
Electrical	
AC Input Frequency Range	47–63 Hz
AC Input Voltage Range	85–264 Vac
Output Voltage	12 Vdc
Output Current	1.0 A min., 8.5 A or 13.0 A max.
Maximum Output Power	102 W or 156 W continuous

Table A-4 STK-RPS-150PS Specifications (continued)

Item	Specification
Physical	
Dimensions	19.6 H x 5.2 W x 25.7 D (cm) 7.7 H x 2.04 W x 10.1 D (in.)
Net Weight (Unit Only)	1.75 kg (3.85 lb)
Gross Weight (Packaged Unit)	3.20 kg (7.04 lb)
MTBF	300,000 Hours

STK-RPS-500PS Specifications

Table A-5 provides the specifications for the STK-RPS-500PS.

Table A-5 STK-RPS-500PS Specifications

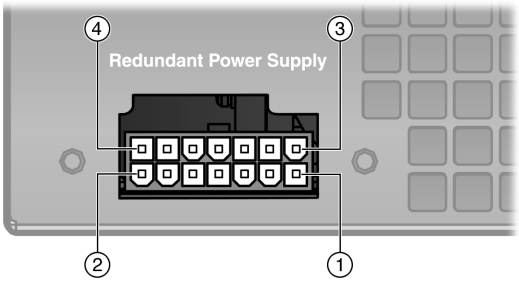
Item	Specification
Electrical	
AC Input Frequency Range	47 - 63 Hz
AC Input Voltage Range	90 - 264 Vac
AC Input Current	10 A maximum at 115 Vac, 60 Hz 5.0 A maximum at 230 Vac, 50 Hz
DC Output	-55 Vdc, 7.5 A maximum +12.0 Vdc, 1.0 A minimum, 10.5 maximum
Maximum Output Power	500 W
Physical	
Dimensions	4.45 H x 44.5 W x 16.5 D (cm) 1.75 H x 17.5 W x 6.5 D (in.)
Net Weight (unit only)	3.47 kg (7.63 lb)
Gross Weight (1 unit and packaging)	4.95 kg (10.89 lb)
MTBF	589,644 Hours at 25°C (77°F)
Environmental	
Operating Temperature	5°C to 40°C (41°F to 104°F)
Storage Temperature	-30°C to 73°C (-22°F to 164°F)
Operating Relative Humidity	10% to 90%

STK-RPS-150PS Redundant Power Supply Connector

For pin location and function, refer to Figure A-1 and Table A-6, respectively.

Note: The following information is for troubleshooting purposes only. For proper operation, do not use any other cable except the RPS cable supplied with the STK-RPS-150PS. This cable is specially designed for this application and meets all necessary regulatory and safety standards. The use of non-approved cables will void your warranty.

Figure A-1 STK-RPS-150PS Redundant Power Supply Connector Pin Locations



1 Pin 1 **2** Pin 7 **3** Pin 8 **4** Pin 14

Table A-6 STK-RPS-150CH2 Redundant Power Supply Connector Pin Functions

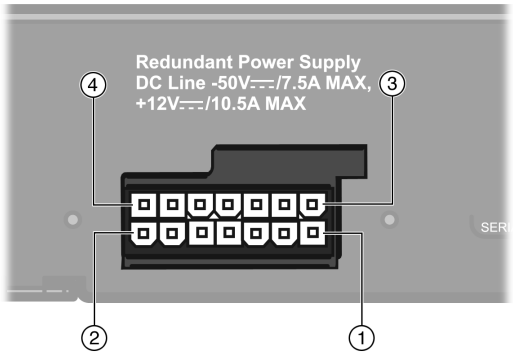
Pin Number	Function	Pin Number	Function
1	Ground	8	Ground
2	No connection	9	No connection
3	12 Vdc Output	10	No connection
4	12 Vdc Output	11	Status 1
5	12 Vdc Output	12	Status 2
6	12 Vdc Output	13	Power good
7	Ground	14	Ground

STK-RPS-500PS Redundant Power Supply Connector

For pin location and function, refer to Figure A-2 and Table A-7, respectively.

Note: The following information is for troubleshooting purposes only. For proper operation, use only the RPS cable supplied with the STK-RPS-500PS. This cable is specially designed for this application and meets all necessary Regulatory and Safety standards. The use of non-approved cables will void your warranty.

Figure A-2 STK-RPS-500PS Redundant Power Supply Connector Pin Locations



1 Pin 1 **2** Pin 7 **3** Pin 8 **4** Pin 14

Table A-7 STK-RPS-500PS Redundant Power Supply Connector Pin Functions

Pin Number	Function	Pin Number	Function
1	- 55 V return	8	Ground
2	- 55 Vdc	9	No connection
3	12 Vdc output	10	RPS present
4	12 Vdc output	11	Status 1
5	12 Vdc output	12	Status 2
6	12 Vdc output	13	Power good
7	Ground	14	Ground

Torque Values

Table A-8 describes the recommended torque values to use when installing the using standard threaded fastener machine screws and bolts.

Table A-8 Recommended Torque Values by Screw Size

Screw Size		Torque in Pounds			Bit Size
English	Metric	-%5	Nominal	+%5	
N/A	N/A	1.42	1.5	1.57	0
2 – 56	1.5	2.85	3.0	3.15	0
4 – 40	2.5	4.75	5.0	5.25	0/1
6 – 32	3.5	8.55	9.0	9.45	1
8 – 32	4.5	17.10	18.0	18.90	2
10 – 32	5	30.40	32.0	33.60	2
1/4 – 20	6.5	63.65	67.0	70.35	3

Pluggable Transceiver Specifications

For SFP and 10G transceiver specifications, refer to the datasheet at http://www.enterasys.com/products/transceivers-ds.pdf.

Console Port Pinout Assignments

Refer back to Figure 2-17 on page 2-23 for information about console port pinout assignments.

Regulatory Compliance

This product meets the safety, electromagnetic compatibility (EMC), and environmental requirements listed in Table A-9.

Table A-9 Compliance Standards

Regulatory Compliance	Standard
Safety	UL 60950-1, FDA 21 CFR 1040.10 and 1040.11, CAN/CSA C22.2 No. 60950-1, EN 60950-1, EN 60825-1, EN 60825-2, IEC 60950-1, 2006/95/EC (Low Voltage Directive)
Electromagnetic Compatibility (EMC)	FCC 47 CFR Part 15 (Class A), ICES-003 (Class A), EN 55022 (Class A), EN 55024, EN 61000-3-2, EN 61000-3-3, AS/NZ CISPR-22 (Class A). VCCI V-3. CNS 13438 (BSMI), 2004/108/EC (EMC Directive)
Environmental	2002/95/EC (RoHS Directive), 2002/96/EC (WEEE Directive), Ministry of Information Order #39 (China RoHS)